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DECENTRALISATION AND QUALITY OF GOVERNMENT

The overlooked role of elite dispersion

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Abstract

The actual effects that decentralisation has on Quality of Government has found no utterly convincing answer in the literature. This thesis seeks to bring clarity into the debate by focusing on a somewhat overlooked core mechanism in the decentralisation literature, elite capture. As such, the thesis seeks to explore whether the dispersion of elites in a country would modify the outcome of different types of decentralisation reforms. To achieve this, two indexes that measure elite dispersion and elite dispersion conditional on elite cohesion are constructed, and the theory is tested by utilising a fixed effects panel regression with a moderated interaction. The thesis concludes that two types of decentralisation are in fact moderated by elite dispersion conditional on elite cohesion, albeit to different effects. While the hypotheses that countries where elites are dispersed and have unified interests would see a lessening of QoG is confirmed in the case for political decentralisation, the opposite seems to be true for fiscal decentralisation

Key words: Decentralisation, Quality of Government, Elite capture, Panel regression, Moderated interaction model

Introduction

“No... In the yacht toilet manager, here captain”

Quote by “Abigail” from the movie Triangle of sadness (2022)¹

A spectrum of literature has linked quality of government to a great deal of essential societal aspects. Among the beneficial effects of high quality of government is increased government legitimacy, economic prosperity and better provision of public goods, to mention just a few (Nistotskaya, 2020). On the other hand, low quality of government in the form of high corruption incurs massive costs in a society, and once corruption has become endemic it becomes an unofficial tax on society, misallocates resources, hinders economic growth, exacerbates economic inequalities and worsens the impact of climate change; among other things (Ades & Di Tella, 1999; Dreher & Herzfeld, 2005; Jain, 2001; Olken & Pande, 2012; Sundström et al., 2026). It is therefore a natural interest for countries to raise QoG and lower corruption, but the question remains on how to do so effectively.

One potential solution that has received much scholarly and policy attention in the last 30 years is decentralisation (e.g., Bardhan, 2002; Bardhan & Mookherjee, 2006a; Bardhan & Mookherjee, 2006b; Fisman & Gatti, 2000; Fjeldstad, 2004). By decentralising, a country can theoretically lower corruption and strengthen the quality of government by bringing government closer to the people (Faguet, 2014; Fisman & Gatti, 2002; Kyriacou & Roca-Sagalés, 2011; Treisman, 2002). The actual effect that decentralisation has on a country’s quality of government is disputed however, and empirical findings scattered. As such, it could be that these reforms in the worst case scenario rather have detrimental than beneficial effects. In order to bring more clarity to this relationship, recent studies have increasingly begun to take different contexts and conditionalities into account, so as to explain what seems to be an evermore complex dynamic. This thesis follows that track, and proposes that an hitherto unresearched factor could explain seemingly divergent outcomes of decentralisation reforms.

¹ Quote taken from the movie’s IMDB page.

More specifically, by focusing on elite capture as the prime mechanism through which decentralisation reforms would be detrimental for quality of government, the thesis examines the moderating effect that elite dispersion has on this relationship. In doing this, the thesis contributes to the field in a number of ways. Firstly by clarifying a conceptual distinction in the literature on elite capture which, to the best of my knowledge, has not been clarified by previous scholars. Secondly, the thesis develops measures of elite dispersion which have so far been absent from the field. Finally, the results of the analyses conducted in the thesis suggest that elite dispersion indeed moderates the effects different types of decentralisation have on the quality of government of a country.

Previous research

This section includes research relevant for the thesis grouped in themes. Firstly, a brief definition of quality of government is presented, after which early research and theories as to how decentralisation would strengthen or weaken quality of government are outlined. As elite capture is a core mechanism in these theories, and of primary interest in this thesis, a section on elite capture follows. Finally, a section on more recent research, with a subsection on the effects of decentralisation when inequalities are taken into account, wraps up the literature review before the theory of the thesis is presented.

Quality of Government

As the name implies, the higher the quality of government (onwards referred to as QoG), the better a society runs. The definition of QoG used in this thesis is that it is the impartiality of how government authority is exercised. For example, low quality of government countries are marked by institutions wherein the same policy is unlawfully applied differently in different situations, while in high QoG countries policy is applied equally between the citizenry unless a certain difference in treatment is stipulated by law (Rothstein, 2021; Rothstein & Teorell, 2008). To be exact, if a person receives different treatment by governmental institutions as compared to their peers because of something like their social standing, that is low QoG. The only exception is if different treatment is codified beforehand, for example a policy that stipulated that single mothers should get prioritised access to government subsidised housing would not be considered low QoG. As such, QoG refers to the norms which guide the way the

government does things, rather than what it does specifically (Bubbico et al., 2017, p. 888 ; Rothstein & Teorell, 2008).

Further, a prevalent hallmark of low QoG is corruption, as one bribing their way into preferential treatment by authorities severely undermines impartial exercise of government authority. In this way the terms are intrinsically linked, and there is this considerable overlap between research that focuses on QoG and that which focuses explicitly on corruption. As such, both research on QoG and on corruption will be of interest in this thesis. It should be noted that QoG goes beyond corruption, however. The concept includes other partial practices which might not be illegal per se, like patronism, nepotism and clientelism to name a few (Rothstein, 2021; Rothstein & Teorell, 2008). Still, the concept of QoG and the phenomena of corruption exist along the same spectrum, with their respective beneficial and detrimental effects for societies.

Decentralisation and corruption/QoG, early research and theories

Decentralisation is by no means a simple term, and, as such, it does not come with a neat one sentence definition. Some scholars (eg. Schneider, 2003; Smoke, 2015, pp. 98–99) make a point of dividing the term along several axes, firstly by stating that decentralisation can refer to either deconcentration, devolution or delegation. Deconcentration refers to central government establishing units on the local level, while devolution refers to the central government transferring certain state functions to lower levels of government, and along with it administrative, economic and political attributes (idib; ; Faguet, 2014, p. 3; Fjeldstad, 2004, p. 1). Finally, delegation refers to the central government contracting another entity to carry out a certain function (Smoke, 2015, p. 98). More importantly, it is further divided into the dimensions of political, fiscal and administrative decentralisation² within the aforementioned divisions. As such, political decentralisation is often referred to as when local governments have elected officials, giving people more direct political means to hold their local government accountable (Bardhan, 2002, p. 186; Seabright, 1996; Smoke, 2015, p. 98; Treisman, 2002). Fiscal decentralisation on the other hand is the transfer of expenditure and revenue power to local government, meaning that local units choose how to spend money as

² Recently, digital decentralisation has emerged as a new dimension. For example in Goel and Saunoris (2016). This dimension will not be considered in the thesis, but could be more relevant for future work.

well as how much to collect in the form of taxes (Enikolopov & Zhuravskaya, 2003; Fisman & Gatti, 2000; Kolstad & Fjeldstad, 2006; Prud'homme, 1995).

Despite a breadth of theorising, research and real world results to lean on, the academic debate on whether decentralisation actually lowers corruption and improves QoG has not found a conclusive answer (Albornoz & Cabrales, 2013, p. 103; Faguet, 2014; Fjeldstad, 2004). The theoretical arguments for decentralisation to lower corruption are manifold, but a core argument made by proponents of decentralisation is simply that it brings government closer to the people who are governed. Local governments are expected to be more knowledgeable about local conditions allowing government to be more efficient, corrupt practices would be harder to hide for local governments, and local officials are more easily held accountable for any wrongdoing as they more easily can be identified and voted out (Bardhan & Mookherjee, 2006b; Faguet, 2014; Fisman & Gatti, 2000; Kolstad & Fjeldstad, 2006; Seabright, 1996; Treisman, 2002). An example of an early influential empirical study pointing in this direction is that of Fisman and Gatti (2000) who, by utilising a cross sectional analysis, find that fiscal decentralisation is indeed correlated with lower rates of corruption.

On the other side of the spectrum are those who argue that decentralisation rather brings about higher corruption rates. These scholars argue that local governments are not only brought closer to their constituents, but also to local elites and special interests, which can more easily “capture” government to serve their interests (Bardhan & Mookherjee, 1999; Prud'homme, 2005; Treisman, 2002). Such arguments gain empirical support in studies like Treisman's (2000), which finds that, along with other factors, federal systems of government are indeed correlated to higher rates of corruption. As such, early empirical studies found support for both a beneficial and detrimental effect of corruption.

Whereas Fisman and Gatti (2000) find that fiscal decentralisation lessens corruption, Treisman (2000) finds that federal systems, which measure political decentralisation, exacerbate the problem. The diverging results could be based on the fact that these are two different types of decentralisation, something that will be further looked into in the section on more recent research, but these early empirical studies nonetheless showcase a divide in the literature that has persisted. It should also be noted that the literature is not limited to the

arguments and mechanisms presented above, and a number of other mechanisms for how decentralisation would impact QoG have been made (reviews include Dawson et al., 2024; Fjeldstad, 2004; Jain, 2001; Kolstad & Fjeldstad, 2006; Treisman, 2002). What these theories and studies highlight however, is the fundamental theoretical role of elites. Whether local governments are more efficient and held accountable to its people to a greater degree, or rather at the mercy of local elites, is at the heart of the question of whether decentralisation is beneficial or not.

Elite capture

Elite capture has been mentioned above as the prime mechanism through which decentralisation would lead to a rise in corruption. Put simply, elite capture is the phenomenon where public resources that would go to the benefit of a broad section of society are being usurped by a powerful few (David-Barrett, 2021; Dutta, 2009, p. 3). Within the literature on elite capture at least two distinct types of elites and elite capture seem to be referred to in different articles. This section will describe both what I would refer to as grand and petty elites, whereas grand elites are of more theoretical interest in this thesis.

Elite capture by grand elites is arguably the most often referred to in decentralisation literature. The characteristics of this type of capture would be that it refers to capture by elites that represent particular, and most typically economic, interests. Grand elites could for example represent a dominant industry, and thus possess a considerable wealth advantage over their peers. Their goal with government capture is to shape government so as to serve their interests, and they do so by building close relationships with political insiders, supporting political candidates that further their interests, and so on. The results of capture can in extreme cases be that the system is wholly reshaped to not only steer resources to the elites' areas of interest but also to widen economic inequalities and disable the checks and balances that would allow the citizenry to rectify the situation (David-Barrett, 2021). An impactful counterargument to decentralisation as a means to lessen corruption has therefore been that local governments are more easily captured by such elites than central ones, meaning that corruption would rather increase as a consequence of decentralisation, when local governments are captured by their local elites (Prud'homme, 1995). However, and as will be further elaborated in the section on the theory of this thesis, whether local

governments are more or less easily captured may be relative and dependent on a few key factors (Bardhan & Mookherjee, 1999; 2000).

Apart from grand elites, another type of capture is carried out by what I refer to as petty elites. Petty elites are those with a comparative power advantage over the rest of the population but who do not represent a particular industry interest. This type is most relevant when capture of development initiatives or aid is studied, where the capture of these exterior resources leads to detrimental effects in villages and small towns, predominantly across developing countries. For example, India is often studied as a country where the social institutions allow local petty elites, in this case someone belonging to a certain caste, gender, etc. to usurp development funds to enrich themselves (Dutta, 2009). Case studies also include other countries, like villages in Indonesia where lack of oversight and accountability can allow village heads to usurp aid that is allocated to the village as poverty alleviation. In extreme cases, they demand bribes for things that are supposed to be given for free, thus excluding the poor families that are supposed to be the primary recipients in favour of those who can afford it as well as supporters of the village head in question (Lucas, 2016).

Although not explicitly conceptualised as different forms of capture in the literature, elite capture is here interpreted as referring to either a system transforming grand capture, in which case the elites responsible are those who represent certain economic interests, or the acts of local petty elites who represent no apparent interest, apart from enriching themselves, usurping external resources such as grants from the central government or other actors. The corruption that is associated with these different types of capture would plausibly also come in different forms, from bribe taking and political-business collusion to “simple” pocketing of public funds. The insight of there being these different types of elites and elite capture is important to keep in mind for the thesis, as this will be relevant both for theoretical clarity as well as when considering measurements.

Recent research on decentralisation and QoG

Lately, research on the relationship between decentralisation and QoG has found the relationship to be complex and dependent on conditionalities and contexts (Bardhan, 2002; Dawson et al., 2024; Smoke, 2015) or as Lessmann and Markwardt (2009) conclude, one size (of decentralisation) does not fit all.

Mainly these studies have focused on the effects of different kinds of decentralisation rather than any effect of decentralisation as a whole. If we first turn to fiscal decentralisation, there seems to be some weight behind the claim that it decreases corruption, as this result is consistently found in a meta-analysis of 71 studies covering three decades of scholarly work. With that being said, there are still a number of studies who do not find empirical support for the claim that fiscal decentralisation lowers corruption. For example, recent country case studies of the US (Shon & Cho, 2019) and Indonesia (Siburian, 2022) have found that fiscal decentralisation in these countries has increased rather than lowered corruption. Perhaps then, the answer to why studies on decentralisation can yield such diverse results may be found in conditionalities and other factors which separate countries from each other. This would explain why decentralisation could seemingly work one way in one study and another way in another, which often include variations in methodology and the countries studied. Along this vein, Bojanic (2023) finds indications that fiscal decentralisation actually worsens corruption, a relationship which is altered when inequality is taken into account however. Likewise, Lessmann and Markwardt (2009) find that the impact of fiscal decentralisation on corruption is unclear unless freedom of the press is taken into account. Kyriacou & Roca-Sagalés (2011) find that while fiscal decentralisation improves government quality, this positive effect is dampened if accompanied by political decentralisation, highlighting that different types of decentralisation can have quite different effects.

In regards to political decentralisation, the field is even more scattered than that of fiscal decentralisation. The meta analysis by Dawson et al. (2024) finds the effect of political decentralisation to be more inconclusive than some studies claim. Political decentralisation's impact on corruption is often found to be dependent on other contextual factors, which has led to the conclusion that in some cases, it may indeed worsen corruption. Fan et al. (2009) find that although political decentralisation in the form of elections at lower levels does not have

an impact on corruption in either direction, possibly due to elites manipulating local elections and/or it requiring more coordination to vote bad or corrupt local officials out of office. A higher number of tiers in a state (i.e. local, regional and national), however, does lead to significantly higher corruption. The authors argue that rather than providing better information on local government and improving effectiveness, multiple tiers muddy the picture, increasing opportunities for bribe extraction, allowing more instances where this can happen and requiring more coordination on behalf of voters for incumbents to be punished at the polls. Albornoz & Cabrales (2013) on the other hand argue that successful political decentralisation is possible, albeit conditional on there being sufficient political competition. According to their model, decentralisation affords voters on the subnational level better information of their politicians as they can compare their region to other regions, and given that there are sufficient political alternatives politicians who perform poorly would be voted out. If political decentralisation and sufficient competition is not in place, however, then decentralisation would lead to more corruption than centralisation.

To summarise, although the relationship between decentralisation and QoG has been studied with evermore complex models taking into account different types of decentralisation, contexts and conditionalities the results of different studies still point in different directions. If elite capture is indeed a core mechanism through which decentralisation would lead to higher rates of corruption, then studies and theories that delve deeper into the factors which enable elite capture are of particular interest. As elite capture relies on a sufficient advantage for the elite over the rest of the population, inequality could be such a variable.

QoG, decentralisation and inequality

Corruption is said to worsen inequality (Gupta, 1998; Li et al., 2000), and, inversely, inequality has been found to worsen corruption (Jong-sung & Khagram, 2005; Policardo & Carrera, 2018). When it comes to the studies that factor in inequality together with decentralisation and QoG, there seems to be two main directions that are taken. Quite a few studies focus on geographical inequality, i.e. between regions, while some but less attention has been paid to interpersonal inequality.

First, when it comes to inequalities between regions, having a large disparity between richer and poorer regions, this has been found to influence the relationship between decentralisation and QoG (Kyriacou et al., 2016; Nguyen et al., 2020). Grounded in previous research of each individual relationship, these two studies provide a framework describing a complex relationship between the three factors in which two-way causal relationships exist between each factor. The authors argue that inequality, decentralisation, and QoG can all influence one another; meaning that they are endogenously set together. For example, there are arguments for decentralisation to both widen or lessen regional inequalities, and for regional inequalities to either increase or decrease the chance that a country decentralises in the first place. This is because centralised states are better able to redistribute resources from wealthier to poorer regions, meaning that decentralisation would lead to a reduction in redistribution and thus widen the gaps between regions. Alternatively, decentralisation affords regions the opportunity to make better informed policies which can lessen disparities by giving poorer regions the freedom to enrich themselves, as well as interregional competition providing the incentive for local governments to do so. As such, the impact of decentralisation could be determined by the level of inequalities between regions that already exist, which, in turn, could also influence the political decision to decentralise or not decentralise in the first place as poorer regions would resist decentralisation and vice versa for richer regions (Kyriacou et al., 2016; Nguyen et al., 2020). In order to test this the authors employ a simultaneous equations model (SEM), which takes into account these types of two-way relationships.

In a study of 23 OECD countries stretching from 1984-2005, Kyriacou et al. (2016) find support for a number of such theorised relationships, albeit not all. They find evidence that fiscal decentralisation lessens regional inequalities and improves QoG, and poor QoG is associated with higher regional inequalities and less fiscal decentralisation. The authors do not, however, find support for the hypothesis that greater regional inequalities would lead to more fiscal decentralisation. The implication of this study is that in order for fiscal decentralisation to reduce regional inequalities, QoG must first be addressed etc. Further, the authors argue that their analysis shows that previous studies suffer from omitted variable bias, as most studies only focus on one of the three relationships while not taking into account the third. This means that it is not sufficient to only look at how fiscal decentralisation impacts regional inequalities, because the result of such a reform would also differ depending on the

level of QoG in the countries (Kyriacou et al., 2016)³. While this study highlights the importance of regional differences for fiscal decentralisation to improve QoG and vice versa, it does so in a convoluted way which blurs the causal chain to a degree that its conclusion risks becoming circular arguments. If a country seeks to improve its QoG and lessen regional inequalities by decentralising, then this might be successful or not depending on its QoG.

Bojanic (2023) on the other hand, studies the moderating effect of interpersonal inequality on decentralisation and QoG, and argues that the effect of decentralisation on corruption is contingent on there being less interpersonal inequality. The study interacts several indicators of decentralisation, fiscal, political and administrative and correlates these with a variety of perception based indices of corruption, using a dataset with both OECD and non-developed countries. The author finds scattered results when only interacting decentralisation and corruption, with results both going against previous research and varying between the group of countries tested, which puts to question the validity of previous research. The study then sees if factoring in inequality could bring some clarity to these null results.

When interacting with the reversed gini index, the results indeed become clearer. Fiscal decentralisation lowers corruption when societies are less unequal as per the gini index, and administrative decentralisation lowers corruption in non-OECD countries when moderated by inequality. Only with political decentralisation are the effects less clear. Further looking at the marginal effects of inequality, the author finds that fiscal decentralisation lowers corruption up to a gini index of around .32, beyond which fiscal decentralisation has a negligible effect on reducing corruption. This result also diverges between groups however, with fiscal decentralisation becoming ineffective in OECD countries for a score of 29, whereas it remains an effective tool in non-OECD countries up to a score of 48-49. The other facets of decentralisation tell a similar story where decentralisation measures combat corruption up to certain levels of inequality, where the tolerance for inequality is higher for non-OECD countries, which is made especially clear with the broad index of decentralisation showing an effectiveness up to a gini index of 29 for OECD countries and 60 for developing ones. It

³ Another point of consideration is that the study by Kyriacou et al. (2016) is limited in that it only includes OECD countries, whereas decentralisation's effect on regional inequalities has been found to vary between developed and developing countries (Lessmann, 2012). This could speak against the generalisability of these results. However, a similar result is reported by Nguyen et al. (2020) in their case study of Vietnam using the same variables.

should be noted, however, that these marginal effects carry with them rather large confidence intervals which often cross both sides of positive and negative correlations, with only the extremes containing confidence to either side of zero (Bojanic, 2023). As such, the somewhat inconclusive results of Bojanic's marginal effects analysis would indicate that interpersonal inequality still does not provide the whole picture, as the effects of this measure becomes negligible around certain values.

Taken together, the studies presented in this section show that interpersonal and territorial inequality are important markers of when and in what contexts decentralisation reforms are effective in the goal of combating corruption and strengthening QoG, but they are not exhaustive in their explanatory power. While Kyriacou et al. (2016) & Nguyen et al. (2020) show that territorial inequalities affect the outcome of decentralisation reforms in countries as a whole, they do not present a clear causal chain nor do they take into account interpersonal inequality, which is where theories of elite capture gain most credence. Similarly, Bojanic (2023) does not take into account territorial factors which have been shown to be relevant for the outcome of decentralisation reforms. Together however, these studies show that both geographical factors as well as interpersonal inequalities matter for the outcome of decentralisation reforms.

A question then arises, what if these are two different facets of the same underlying phenomena? The theory of this thesis explores precisely such a possibility, by focusing on both geographical qualities of countries as well as their interpersonal inequalities, which are both seen through the lens of the core theoretical mechanism of elite capture.

How variation in elite dispersion could affect capture rates and QoG

This thesis sets out to test whether a previously unresearched variable, the geographic dispersion of elites in a country, could modify the impact of decentralisation on elite capture of local governments, and thus QoG. That this could be the case seems a logical but overlooked next step given some of the previous theory and research outlined above, which for the most part only makes brief mentions of what is arguably a core theoretical dynamic. By focusing on elite capture more explicitly, the thesis might hopefully bring some clarity into the scattered results that have marked the field as a whole.

The theoretical expectations of the thesis largely come from the theory of relative capture of local governments developed by Bardhan and Mookherjee (1999; 2000). The theoretical framework provided by these authors delves into elite capture more specifically, and in doing so they describe a dynamic which can be used to explain not only under which circumstances elite capture of local governments is a higher risk, but also how the development of countries as a whole can be understood. Their formal model seeks to explain relative levels of political capture between the central and subnational levels of government by describing the factors that determine such an outcome.

The model is made up of voter awareness, electoral uncertainty, the electoral system, interpersonal inequality in the district and interest group cohesiveness. Variations in these factors make the outcome of this equation uncertain, meaning that local governments can be either more or less easily captured than a central one depending on context. For example, poor, unaware and undecided voters together with fewer separate elite interests (i.e., higher interest group cohesiveness) at a local level will make these governments relatively easier to capture, especially when the inverse is true on the central level. Unlike previous theories which simply state that decentralisation would lead to higher rates of elite capture (eg. Prud'homme, 1995), Bardhan and Mookherjee show that easier capture of local government is not a given, but something that will vary based on differences in elite power as compared to the rest of the population.

The theory of this thesis extrapolates the logic of such a dynamic, imagining what would be expected as it is applied to countries as a whole. Before doing so however, some clarifications should be made. Firstly, the theory of relative capture is limited to one form of decentralisation in that it only focuses on elite capture of elections, primarily by use of lobbying. As such it does not include further clandestine activities that are of interest in this thesis such as bribery or capture under other forms of decentralisation. Secondly, it should be noted that only one of the types of elites identified in the section on elite capture is of main theoretical interest, namely grand elites. Not only are they consistently the type of elite in focus when theories warn of the dangers “local interest groups” entail for local governments (eg. Prud’homme, 1995, pp. 211-212; Bardhan, 2002, pp. 192–193; Bardhan & Mookherjee, 1999; 2000) and as such of most interest for the field as a whole, but they are more easily identified. While petty elites can potentially be found anywhere, specific economic interests and grand elites are not uniform across a country. Grand elites will differ in presence between countries, being dispersed across most regions in some countries, or concentrated in a few regions in others. As such the thesis assumes that the dynamic of relative power of grand elites described by Bardhan and Mookherjee (1999; 2000) would entail noticeable and measurable differences between countries. Given that decentralisation enables elite capture of local government, differences in how dispersed grand elites are in a country would then lead to different outcomes in overall capture rates between countries.

To illustrate this, imagine two countries where all else is equal apart from elite presence and dominance. Each country has 10 regions, but in the first country grand elites are both present and sufficiently powerful enough compared to the rest of the population to capture local governments in 9 out of 10 regions. In the second country, however, grand elites are not as uniformly powerful across regions, and are therefore only able to capture 1 out of 10 regions. When both countries decentralise, the first country would see a capture rate of local governments in 9 out of 10 regions, whereas the capture rate in the second country would only be 1 out of 10 regions. Since elite capture of local governments is detrimental to QoG, the first country can be expected to see a decline in QoG as a result of decentralisation, whereas the second country would not⁴.

⁴ As there are a multitude of theorised positive and negative aspects of decentralisation (these are outlined well in Treisman (2002)), the effect a decentralisation reform ends up having could be dependent on which aspects outweigh others. For example, if elite capture is more detrimental for QoG than the benefit of local governments

As such, the first hypothesis of the thesis is:

H1: A country with more dispersed grand elites will suffer greater capture rates, and lower QoG, with decentralisation.

Given the factor of elite cohesion, however, it may be that grand elite presence in itself might not be sufficient for capture of local governments. As another requisite outlined by Bardhan and Mookherjee (1999; 2000) is interest group cohesiveness, it could be that grand elites also have to be unified in interest for capture to take place. If grand elites are not unified in interest they could instead end up competing with each other. For example, if a region has grand elites where some represent industrial interests and others agricultural ones, it is reasonable to assume that they would have different policy preferences and therefore lobby for different candidates etc. Such an environment would not easily permit the type of grand elite capture described by David-Barrett (2021) in which the government is more or less completely reoriented to serve a sole economic interest. As such, countries with this type of environment in its regions could avoid a large downturn in QoG, despite decentralising while having grand elites present in many regions. The second hypothesis therefore goes beyond H1, specifying that elite presence is further conditional on cohesiveness:

H2: A country with dispersed and unified elites will suffer greater capture rates, and thus lower QoG, with higher decentralisation.

Finally, as has been touched upon in a number of studies: not all decentralisation is equal. A recurring theme in theories of elite capture is that grand elite capture seems to be most associated with politics and thus political decentralisation. This is evident in both the mechanisms of elite capture as described by David-Barrett (2021) who describes close relationships between grand elites and politicians as a main avenue of elite capture. It is also evident in theories such as that of Bardhan and Mookherjee (1999; 2000) described above, which focuses on the relative susceptibility of local government capture by elite interests in

having better knowledge, then the effect of decentralisation is nonetheless detrimental. In countries like the second country in this theoretical example however, where capture of local governments is not widespread, the beneficial effects of decentralisation would be abundant. As such, the differences between countries with dispersed and concentrated elites would be even more stark, as the countries without dispersed elites would reap mostly positive effects of decentralisation whereas the QoG in those with dispersed elites would only suffer from it.

elections. Inversely, elite capture does not seem to be as dominant of a mechanic in other forms of decentralisation. As such, the third hypothesis is:

H3: Political decentralisation is particularly sensitive to elite capture.

Method

The methods section will first outline the construction of two indexes which measure elite dispersion and elite dispersion conditional on cohesiveness, respectively. After this, the model used to test the hypotheses is outlined, as well as the data that is used to operationalise dependent, independent and control variables.

Elite dispersion index

A major obstacle to overcome in this research design is how to measure elite dispersion. To the best of my knowledge, no such measure exists, which means that the measure has to be constructed from scratch in order to answer the hypotheses. This elite dispersion index, or EDI as I will call it, should therefore be able to measure how spread out grand elites are in one country compared to another.

The EDI is based on data from Chrisendo et al. (2024), who have been able to construct a regional gini index which runs through the years 1990-2021. To be specific, the data used is that of the World Inequality Database (WID). The reasoning behind using gini coefficients is that it is a widely used measure of wealth inequality, which according to the theory outlined above is a main determinant of grand elite capture. Wealth inequality largely determines not only whether a grand elite is present, who by definition hold a considerable advantage in wealth as compared to the rest of the population, but also enables grand elites to capture government by paying lobbyists and/or bribing key officials.

Using these regional coefficients, the index begins by assigning a 1 to each region whose gini score is above the third quartile as per the global data⁵. If the region does not meet this requirement, it is assigned a value of 0. As such a region is defined as having a grand elite presence if:

⁵ The third quartile of regional gini has a value of .581 in the dataset, whereas the mean is $\approx .49$

$$E_{i,c,t} = 1 \text{ if } Gini_{i,c,t} \geq Q3$$

Where:

$E_{i,c,t}$ is the grand elite presence of region i in country c at time t .

$Gini_{i,c,t}$ is the gini coefficient of region i in country c at time t .

$Q3$ is the third quartile, or the regional gini coefficient that is higher than 75% all regional gini coefficients in the dataset.

The mean of elite presence values for each country is then calculated giving the EDI:

$$EDI_{c,t} = \frac{1}{n_{c,t}} \sum_{i=1}^{n_{c,t}} E_{i,c,t}$$

Where:

$EDI_{c,t}$ is the Elite Dispersion Index of country c at time t .

$n_{c,t}$ is the number of regions in country c at time t .

As such, in accordance with H1, a country of 10 regions with grand elite presence in 9 of them would get an EDI of 0,9 while a country with grand elite presence in just one region would get an EDI of 0,1. A problem arises in that small countries could bias the EDI, however. A country of only one region could potentially get the highest possible EDI of 1 although grand elites are very far from dispersed. To rectify this, all countries where $n_{c,t} < 5$ are dropped from the dataset.

Table 1 shows the top 10 countries with the most dispersed elites according to the EDI. The scores shown are the mean of each country over the 30-odd years that the index covers. The fact that all countries apart from one has a value of 1 is thus quite remarkable as this means that every single region of these countries has been among the 25% most unequal for that entire time period. It might be said that the index still biases smaller countries due to many of the listed countries being smaller, for example Qatar, Honduras, Bahamas etc. This will have to be taken into account when choosing the appropriate model for the analyses.

Table 1: Countries with the highest EDI scores

1.	Qatar	1	6.	Equatorial Guinea	1
2.	Eswatini	1	7.	Somalia	1
3.	Suriname	1	8.	Venezuela	1
4.	Honduras	1	9.	Guyana	1
5.	Bahamas	1	10.	Gabon	.9803922

EDI limitations

The EDI brings with it a number of caveats, and it is not necessarily a definitive reflection of actual elite dispersion. For one, it only includes a proxy measurement of one source of imbalance between elites and the general population, namely that of wealth. It does not account for other factors that could also be theoretically relevant, such as personal connections between businesses and politicians or support by key parts of the electorate (David-Barrett, 2021; Lucas, 2016), not to mention actual sustained grand elite presence in the form of residence in the region. The reason that such indicators are not included is that measurements are not available. Mapping out residence and political or other relevant social connections is an undertaking that is far beyond the scope of the thesis, especially as the analysis stretches over a large number of countries, regions and years. This limitation also means that out of the types of elites identified in the section on elite capture, only one type can be said to be measured, namely those whose advantage is derived from wealth, thus omitting the type of capture carried out by local elites who rely on other types of imbalances in power to usurp resources. As only grand elites are of interest to the study, however, this limitation should not have a meaningful negative impact on the study.

Further, the threshold of the Gini index of the third quartile is set ad hoc for the purpose of this thesis and not a number supported by previous research as the level of inequality where elite capture simply happens. To the best of my knowledge, such a number does not exist, and the threshold is therefore chosen on the basis of being a theoretically plausible threshold for

capture due to its magnitude. Nonetheless, tests with different thresholds will therefore be prudent so as to test the robustness of any results.

Another limitation is that since gini coefficients used in the EDI are based on the regional level, the measure does not fully reflect the local imbalances that could lead to the capture of lower levels of government below the regional level. This limitation is important to acknowledge, as not only do local governments outnumber regional governments, but the authority between the two subnational levels of government may also differ. While in federal states the region or subnational state may indeed be a highly relevant object of capture for local elites, other countries may be organised as such that lower levels like municipalities are delegated more responsibilities. This limitation means that the EDI doesn't reflect nor take into account the full spectrum and nuance of subnational governments and elites, which could be of interest to future research.

Finally, while EDI as constructed above provides a measurement for elite presence in the form of disparities between grand elites and general population in wealth, it does not reflect whether elites are unified in interest. As mentioned in the theory section, elite cohesion is arguably an essential determinant of elite capture, as an incoherent elite might be competing too much within itself to successfully capture government. To account for this, I build a second measure of elite dispersion that incorporates a measure of economic diversity into the EDI.

Interest Cohesion Elite Dispersion Index

This version of the EDI, called the Interest Cohesion Elite Dispersion Index, is constructed so as to answer H2. If grand elites have to be not only present but also share common interests in order to capture local governments, then the EDI is not sufficient. In order to add a measure of cohesive elite interests, data on economic sector diversification is gathered from the DOSE dataset (Wenz et al., 2023). The dataset gives the share of the sectors manufacturing, agriculture, service⁶ and other in each region, which are used to construct a Herfindahl-Hirschman Index (HHI). The HHI gives a measure of the concentration of the

⁶ The share of manufacturing, agriculture and services are less than 1. As such a fourth, "other", sector is added by calculating $1 - \text{the sum of these sectors}$.

sectors and is often used to detect oligopolies and non-competitive markets (Rhoades, 1993). For the purpose of this thesis, the HHI is used to give a measure as to whether one particular sector dominates in a region, as the index gives a value of 0.25 if a region is as economically diversified as possible, and a value of 1 if only one sector is present. The HHI is calculated as follows:

$$HHI = \sum_{j=1}^4 (j \text{ share}_{i,c,t})^2$$

Where:

j is the sector

$j \text{ share}$ is the share of sector j in region i of country c at time t

This measure is integrated into ICEDI by having the assignment of a value of 1 in a region also be conditional on the second quartile of the HHI⁷, which gives a measure of whether a region has a grand elite presence *and* a concentrated economic sector:

$$UE_{i,c,t} = 1 \text{ if } Gini_{i,c,t} \geq Q3 \cap HHI_{i,c,t} \geq Q2$$

Where:

$UE_{i,c,t}$ is the unified elite presence of region i in country c at time t

$HHI_{i,c,t}$ is the HHI value of region i in country c at time t

$Q2$ is the second quartile HHI values across all regions in the dataset

The definition of the complete ICEDI is therefore:

$$ICEDI = \frac{1}{n_{c,t}} \sum_{i=1}^{n_{c,t}} UE_{i,c,t}$$

Table 2 shows the top 10 countries with the highest average ICEDI scores for the years covered by the index. One thing that stands out is that the spread in scores is much more prevalent than was the case for the EDI, with the country that holds the largest score (Mexico) being a real outlier compared to the rest. The difference between Mexico and Chile per the

⁷ Q2 of the HHI is $\approx .386$ and the mean is $\approx .413$

ICEDI is a whole 0,451 points, and going from there the points drop dramatically as the ICEDI progresses. This will have to be taken into account when the ICEDI is used for analysis.

Table 2: Countries with the highest ICEDI scores

1.	Mexico	.828125	6.	Panama	.1633333
2.	Chile	.3764706	7.	Bolivia	.1182796
3.	South Africa	.3580247	8.	Thailand	.0748297
4.	Brazil	.3398693	9.	Argentina	.0713602
5.	Colombia	.240779	10.	Ethiopia	.0666667

ICEDI limitations

The ICEDI also has its shortcomings. Most blaringly, while accounting for sector domination, it does not reflect true economic diversity in the sense that dominance or diversity of companies are not factored in. A sector like services could include a variety of companies that compete with each other and where elite interests are not aligned, for example, meaning that although the region would be given a high value in the ICEDI, it may still have high competition among grand elites albeit in the same broad sector. For example, a region can be dominated by agriculture, but the interests of owners of massive cattle farms might diverge from those who own massive soy farms etc. This shortcoming is again due to a limitation of data, but if it can be assumed that most regions where a singular sector dominates would give grand elites more similar interests than one where multiple sectors are present, and as such this drawback does not fully interrupt what the ICEDI represents theoretically.

A final shortcoming which is made obvious by table 3 below, is that the regional gini dataset and the DOGE dataset which makes up the HHI have significant gaps in the data between them, which limits the data to much fewer countries. However, although the ICEDI is limited to few countries, as can be seen in table 2 the countries covered are from different parts of the world. Therefore the index should largely avoid the bias that comes with other measurements which only cover OECD countries, for example. Meaning that even though wealthier countries are overrepresented, developing countries are not entirely excluded either.

Comparing the two indexes

Table 3 shows the difference in coverage between the EDI and ICEDI. Comparing the two indexes, a few things stand out. For one the ECI is much more extensive, and covers many more countries as well as a slightly larger timeframe. The means of the different indexes are also quite different, showing that the ICEDI is much more “selective” as less regions are flagged with a 1 due to the index’s more stringent demands, as compared to the high scores of the countries with the highest EDIs. This is further seen in the correlation between the two indexes which is quite low, meaning that the inclusion of cohesive economic interests into ICEDI makes for a different measurement than the basic EDI.

Table 3: Comparison between the EDI and ICEDI

Coverage Summary: EDI vs ICEDI

	mean	sd	min	max	Observations	Number of countries	Year span
EDI	0.295	0.374	0.00	1.00	7008	207	1990 - 2023
ICEDI	0.068	0.170	0.00	1.00	651	45	1990 - 2020
Correlation between EDI and ICEDI	0.3433						

Empirical model

The hypotheses are tested using a fixed effects panel regression with a moderated interaction. A model that uses panel data brings with it several benefits. For one, it focuses exclusively on variation within countries as opposed to between countries, which eliminates bias on the part of time invariant factors that are not accounted for (Bröderl & Ludwig, 2015). Using a model which automatically controls for such omitted variable bias is desirable as some scholars (eg. Lessmann, 2009; Kyriacou et al., 2016) point out that this has been a flaw in many previous studies. Time invariant factors that the model accounts for which are relevant in this context are for example culture, traditions and geographical aspects such as land size. In this study in particular land size is of especial importance since, as was pointed out in the discussions on the construction of the EDI and ICEDI, countries with fewer regions can be much more volatile in their index scores. This is partially mitigated by the indexes excluding countries with less than five regions, and is now additionally accounted for by using models focused on

within unit variation, which should provide a complementary safeguard for potential small country bias. The model also utilises year fixed effects which should account for global shocks like the 2008 financial crisis or the covid epidemic, and all variables except the dependent variable are lagged so as to mitigate simultaneity bias and better estimate the direction of causality. The model is defined as:

$$QoG_{c,t} = \beta_0 + \beta_1 Dec_{c,t-l} + \beta_2 Elites_{c,t-l} + (Dec. \times Elites)_{c,t-l} + \sum ctrl_{c,t-l} + \alpha_c + \delta_c + \epsilon_{c,t}$$

Where:

$QoG_{c,t}$ is the Quality of Government of country c at time t

$Dec_{c,t-l}$ is the form of decentralisation (fiscal, political or overall) in country c at time t minus lag

$Elites_{c,t-l}$ is the elite presence according to either the EDI or ICEDI depending on model

$Dec. \times Elites$ is the interaction term

$\sum ctrl_{c,t-l}$ is the sum of control variables for country c at time t minus lag

α_c is the country fixed effects of country c , which controls for time invariant factors such as culture, size etc.

δ_c are the year fixed effects of country c which controls for external chocks

$\epsilon_{c,t}$ is the residual of country c at time t

It should be noted that when looking at the relationship between three factors such as the ones in this thesis, different studies have taken different approaches. Since some envision relationships between similar variables to simultaneously impact each other, they employ simultaneous interaction models (Kyriacou et al., 2016; Nguyen et al., 2020), while others such as Bojanic (2023) see inequality as interacting with the effect of decentralisation on QoG rather than all factors simultaneously interacting with each other. As specified above, this thesis likewise uses a moderated interaction model.

The methodological and theoretical foundations for using a SEM are still important to acknowledge, however, as interregional inequality could introduce a bias that the current

model does not address as effectively. Previous scholars have argued that a SEM would be a more prudent model based on the theoretical expectation that interregional inequality would affect whether or not countries decentralise in the first place. As theory suggests that the larger the gap between regions the larger the incentive for rich regions to push for decentralisation while poorer regions get the inverse incentive of centralisation (Beramendi, 2007). As such, the rate of decentralisation in a country would be influenced by the territorial inequality of that country prior to decentralisation, while the disparity of territorial inequality would be influenced by the degree of decentralisation. This sort of simultaneous relationship is then argued to necessitate the use of a SEM (Kyriacou et al., 2016; Nguyen et al., 2020).

Still, a moderated interaction model is used rather than a SEM because previous studies which do use a SEM have not found empirical evidence supporting the claim that inequality significantly influences decentralisation in the first place (Kyriacou et al., 2016), reducing the risk of this being a factor that is essential to take into account. Another reason is that while the thesis does incorporate a territorial aspect that focuses on interpersonal inequality, the theoretical foundations of the EDI and ICEDI should not be conflated with those of the regional inequality literature. While there are logically reasons for these two to be correlated to a higher or lower degree, as a region with a high concentration of elites is plausibly also one of the wealthier regions, the elite dispersion measures would logically carry with them a greater degree of elasticity. As such, a potential relationship between elite concentration and incentives to decentralise are not as straightforward as the theorised relationship between regional inequalities and decentralisation. It is plausible for elites to be present in both rich and poor regions, and as such elite concentration does not necessarily carry with it the same incentives as regional inequality. Primarily however, the reason is that a moderated interaction model is in greater accordance with the research question and the theoretical expectations outlined above. The question that the thesis sets to answer is not whether and how much these factors intervene, but rather how the degree of elite concentration affects the outcome of decentralisation. A moderated interaction model is better suited for this task, and reduces the risk that no clear direction of causality can be established.

The discussion above raises the salient problem of endogeneity however, an issue that several studies similar to this thesis acknowledge (e.g. Bojanic, 2023; Bojanic & Collins, 2019;

Enikolopov & Zhuravskaya, 2003; Kyriacou et al., 2016; Kyriacou & Roca-Sagalés, 2011; Lessmann, 2009; 2012; Nguyen et al., 2020) and many also employ lagged variables, among other measures, to account for it (Bojanic, 2023; Bojanic & Collins, 2019; Enikolopov & Zhuravskaya, 2003; Kyriacou et al., 2016; Kyriacou & Roca-Sagalés, 2011; Lessmann, 2009; 2012; Nguyen et al., 2020). However, Lessman (2012) warns of this as being an insufficient measure. Since the problem lies in the endogenous variable being correlated with the error term, lagging decentralisation as an instrument does not rid the regression of endogeneity as decentralisation is persistent and does not vary often, meaning that the correlation between the error term and lagged decentralisation is difficult to account for with lags. The thesis addresses this problem by utilising incremental time lags (3 years in the main models, with 0,1 and 5 year lags being used to test robustness) with the logic being that alternating configurations would expose whether lagging is indeed an insufficient measure. If this is the case it could cast doubts on the validity of the chosen model, and as such future studies utilising more statistical models like a SEM could be prudent.

Data

For the independent variables the study utilises measurements established in previous studies, and fiscal decentralisation is operationalised as the subnational share of expenditure or revenue as suggested by Schneider (2003) in his article on decentralisation measurements⁸. In this case, it is measured as the combined share of subnational and local expenditure/revenues from the International Monetary Fund database (2025)⁹. Political decentralisation is in many cases measured as a simple binary of whether a country has subnational governments or not (eg. Fisman & Gatti, 2000; Lessmann & Markwardt, 2009), or in some studies also measured as the number of tiers of a country (Fan et al., 2009; Lessmann & Markwardt, 2009). As pointed out by Lessmann (2009, p. 4) however, such de jure measures can be unsatisfactory in that it does not necessarily reflect real devolution of power to subnational units and thus actual decentralisation. Therefore political decentralisation is operationalised as the variable of “independent sub-federal units” found in the QoG dataset. Independent sub-federal units measure whether a country has sub-federal units that also substantively constrain national

⁸ Examples of studies that follow this measure are (eg. Bojanic & Collins, 2019; Fisman & Gatti, 2000; Lessmann, 2009; Lessmann, 2012)

⁹ Expenditure and revenue decentralisation are calculated as $1 - \text{Central government revenues}$ so as to cover both subnational and local government expenditure/revenue.

fiscal policy (Henisz, 2002; Teorell et al., 2025). As such, the variable not only measures de jure political decentralisation but also the magnitude of such decentralisation. Finally, an overall decentralisation variable is taken from the Regional Authority Index (RAI), which gives an annual aggregated score per country based on the authority in shared rule and self rule wielded by regional governments (Shair-Rosenfield et al., 2020; Hooghe et al., 2016). In the moderated interaction models, these are all interacted with the ECI and EDECI outlined above. It should also be mentioned that the independent and moderating variables are not significantly correlated. As such, multicollinearity should not represent an issue for the analysis. This also means that elite dispersion and the types of decentralisation of interest are distinct political dimensions, which further highlights the relevance of an interaction model.

While many studies use corruption perception indexes or surveys (eg. Bojanic, 2023; Fan et al., 2009; Fisman & Gatti, 2000), variables based on perceived corruption are not ideal for precisely measuring the decline in QoG that would result from grand elite capture. Apart from corruption being inherently difficult to measure due to its clandestine nature (Jain, 2001). Many measures of perceived corruption also do not discriminate on the type of corruption that is perceived meaning that it could refer to anything from “petty” bribe taking to the larger corruption scandals of interest, making them rather inaccurate for specific types of corruption like that of interest in this thesis. This study instead follows Kyriacou et al. (2016) in using the ICRG indicator of Quality of Government as the dependent variable. The ICRG indicator is a mean of law and order, bureaucratic quality and corruption scores, and can be found in the QoG dataset (The PRS Group et al., 2025; Teorell et al., 2025). The ICRG indicator will hopefully provide a more complete reflection of the detrimental effects of state capture on QoG as the measurement seeks to go beyond conception of corruption in general as “...it is more concerned with actual or potential corruption in the form of excessive patronage, nepotism, job reservations, 'favor-for-favors', secret party funding, and suspiciously close ties between politics and business.” (Teorell et al., 2025, p. 1135).

Control variables

Concerning control variables, previous research has identified an abundance of relevant factors. First of these is economic development as measured by logged GDP per capita (Fan et al., 2009; Fisman & Gatti, 2000; Kyriacou et al., 2016; Kyriacou & Roca-Sagalés, 2011) as

GDP per capita has been proven to strongly correlate with corruption, perhaps due to wealth increasing governance through depersonalised relationships and better education (Alonso & Garcimartín, 2013; Treisman, 2000). GDP per capita as per the 2015 dollar value is taken from the World bank data found in the QoG dataset (World Bank, 2024; Teorell et al., 2025) and subsequently expressed in logs. Population size is likewise included as it has been found to determine corruption levels (Fisman & Gatti, 2000; Treisman, 2000). This data is also gathered from the World Bank (2024) data included in the QoG dataset (Teorell et al., 2025).

Trade openness is also often controlled as more open economies would be less corrupt and better governed, mostly following the logic that increased competition undermines rent-seeking behaviour (Fan et al., 2009; Fisman & Gatti, 2000; Kyriacou et al., 2016; Lessmann, 2009; Lessmann, 2012)¹⁰. This thesis follows Lessman (2009; 2012) in controlling for openness as the sum of imports and exports divided by GDP, in this thesis gathered from the QoG dataset (World Bank, 2024; Teorell et al., 2025)

Democracy and freedom of press are controlled for as they have significant effects on corruption (Chowdhury, 2004; Lessmann & Markwardt, 2009; Treisman, 2000). Democracy is operationalised by the Boix et al. (2012) dichotomous democracy measure found in the QoG dataset (Teorell et al., 2025). Press freedom is operationalised using the Freedom House (2017) index in the QoG dataset¹¹ (Teorell et al., 2025).

Finally, as in other studies, ethnic fragmentation¹² (Enikolopov & Zhuravskaya, 2003; Kyriacou & Roca-Sagalés, 2011; Kyriacou et al., 2016) and share of protestants (eg. Bojanic, 2023; Fan et al., 2009; Kyriacou et al., 2016; Kyriacou & Roca-Sagalés, 2011) are controlled for due to their correlation with corruption. Ethnic fragmentation is operationalised using the dataset by Fearon (2003), which can be found in the QoG dataset (Teorell et al., 2025). The share of protestants is gathered from the World Religion project, national religion database

¹⁰ Although Gatti (2004) suggests that it is the policies that restrict trade that affect corruption, and not openness per se.

¹¹ The data in the QoG dataset is split along three variables which together stretch from 1993-2016, these were added together to form a uniform variable and the values from 1993-1990 as well as from 2016-2023 were carried forward/backward to make up for the missing years. This means that the values are not necessarily representative during those years.

¹² Ethnic fragmentation is also a constant value that will not be included in the FE models, but is indirectly controlled for by the model.

(Maoz & Henderson, 2023). Because it doesn't cover 2010 and onwards, the value for 2010 is carried forward to cover this gap, which should suffice as such demographics do not change much.

The moderators, primarily the EDI but to some extent also the ICEDI, could be said to be nothing but proxies for overall inequality. After all, a country with great inequality overall can be expected to have large regional inequality, meaning that the study risks more or less testing the same hypothesis as Bojanic (2023). To avoid this, national gini coefficients from the same WID data as the subnational gini coefficients (Chrisendo et al., 2024) are also used as a control. A summary of all variables used can be found in table 4.

Table 4: Descriptive statistics

	count	mean	sd	min	max
Quality of Government (ICRG)	4630	0.547	0.213	0.042	1.000
Expenditure Decentralization	984	0.459	0.224	0.002	0.866
Revenue Decentralization	1077	0.360	0.190	0.018	0.909
Independent Subnational Units (h_f)	4959	0.066	0.248	0.000	1.000
Regional Authority Index (RAI)	2673	9.797	9.770	0.000	37.722
EDI	7008	0.295	0.374	0.00	1.00
ICEDI	651	0.068	0.170	0.00	1.00
GDP per Capita (log)	5988	8.437	1.450	5.243	12.227
Trade (% of GDP)	5310	83.355	49.997	0.021	437.327
Protestant Share (%)	4445	0.135	0.205	0.000	0.990
Total Population (log)	6282	15.460	2.213	9.125	21.072
Ethnic Fragmentation	6418	0.439	0.255	0.000	0.930
Democracy (Binary)	5937	0.572	0.493	0.000	1.000
Press Freedom	6612	53.101	24.357	0.000	95.000

Note: Statistics show all available observations per variable.

Empirical results

The results will be presented as they pertain to each independent variable. The tables include bivariate models between just the dependent and independent variable and models with and without controls as a benchmark to compare the interaction models to, as well as interaction models with and without controls for either moderator. The main tables presented are those where independent, moderator and control variables have been lagged three years. Apart from dealing with endogeneity, three years specifically is chosen as it is assumed that a decentralisation reform would likely take more than a year to have a substantial effect on

QoG, and a larger lag would lead to excessive data loss. Tables of the main models without timelag, a 1 year timelag and 5 year timelag can be found in the appendix. Robustness tests in the form of stricter thresholds for the EDI and ICEDI can also be found in the appendix

Expenditure decentralisation

Table 5 shows the models which include expenditure decentralisation as the independent variable. Surprisingly, the bivariate regressions show no significant correlation, which would be expected given that many previous studies find significant positive correlations between fiscal decentralisation and corruption (Dawson et al., 2024). As for the effect of the moderators, while the EDI shows no significant result but a negative coefficient, the ICEDI has a positive and highly significant coefficient when interacting with expenditure decentralisation. This goes directly contrary to H2, which stipulates that a country with dispersed elites unified in interest would lead to a greater degree of elite capture and thus a worsening of QoG. Instead, the result shows that dispersed and unified elites are rather beneficial for QoG as a country decentralises fiscally. It should also be noted that the ICEDI by itself has a highly significant negative coefficient in the same model. This means that when fiscal decentralisation is at zero, a high ICEDI is detrimental for the country in question, a damage which is then alleviated as countries decentralise fiscally.

Graph 1 shows a marginal effects plot of this relationship with quite uncertain results. The confidence intervals remain on both sides of zero through each level of the ICEDI, meaning that the marginal effect of the interaction is insignificant. This reinforces the plausibility of the interaction describing a dynamic where a country with dispersed and unified elites might not necessarily see high QoG after fiscal decentralisation, but rather less poor QoG.

Table 5: Expenditure decentralisation models

Main FE Models: expdec (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
Expenditure decentralisation	-0.005	-0.010	0.012	0.003	-0.077	-0.066
	(0.041)	(0.042)	(0.036)	(0.049)	(0.103)	(0.101)
Expenditure decentralisation##EDI			-0.154	-0.135		

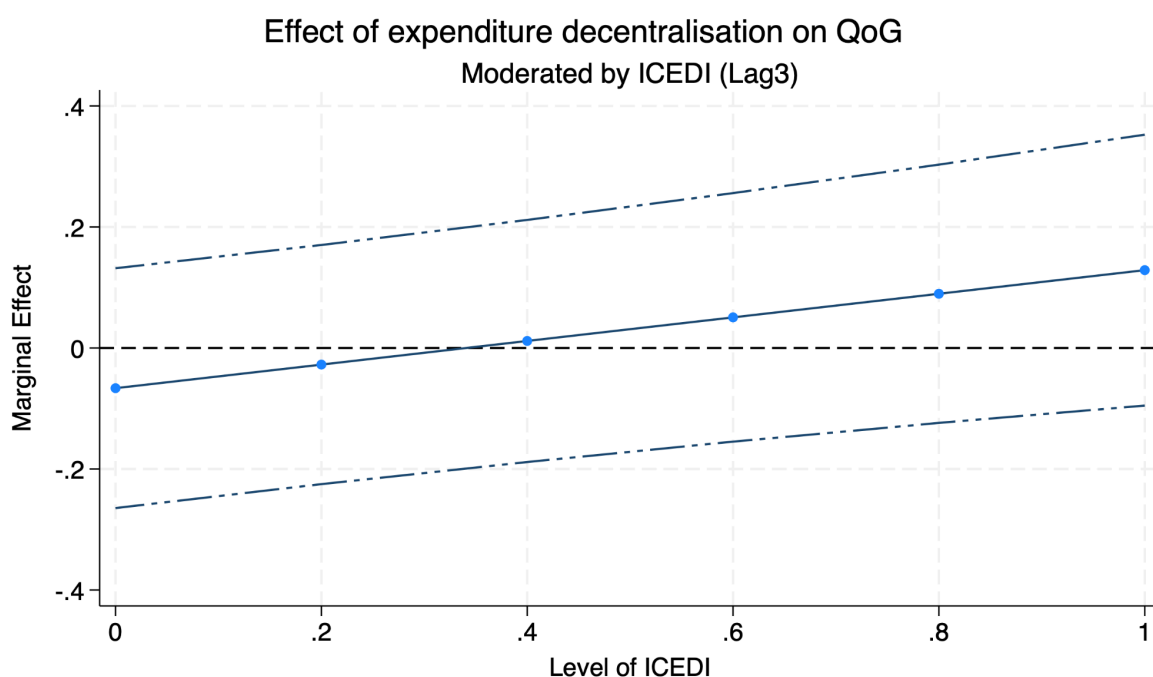
			(0.138)	(0.139)		
Expenditure decentralisation##ICEDI					0.007	0.195***
					(0.063)	(0.062)
EDI			0.076	0.122		
			(0.079)	(0.083)		
ICEDI					0.005	-0.121***
					(0.039)	(0.040)
GDP logged		-0.003		0.003		0.034
		(0.036)		(0.036)		(0.048)
Population logged		0.010		0.000		-0.096
		(0.092)		(0.086)		(0.106)
Trade		-0.001**		-0.001**		0.000
		(0.000)		(0.000)		(0.001)
National gini		-0.030		-0.197		0.127
		(0.257)		(0.260)		(0.153)
Share of protestants		0.016		0.000		-0.384
		(0.113)		(0.110)		(0.398)
Democracy		0.024**		0.021*		-0.082**
		(0.012)		(0.012)		(0.038)
Press freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	689	431	689	431	196	112

Standard errors in parentheses

Thresholds: Base. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Graph 1: Interaction between expenditure decentralisation and QoG moderated by ICEDI, FE.



Revenue decentralisation

As in table 5, the benchmark regressions in table 6 show no significant correlations, but this time neither do the interactions. This means that the models can not discern any effects of either revenue decentralisation, the EDI and ICEDI nor their interactions with revenue decentralisation on QoG.

Table 6: Revenue decentralisation models

Main FE Models: revdec (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
Revenue decentralisation	0.047	0.129	0.035	0.111	0.014	-0.133
	(0.078)	(0.119)	(0.081)	(0.156)	(0.190)	(0.162)
Revenue decentralisation##EDI			0.057	-0.006		
			(0.201)	(0.231)		
Revenue decentralisation##ICEDI					-0.257	0.191
					(0.211)	(0.185)
EDI			-0.016	0.066		

			(0.057)	(0.083)		
ICEDI					0.091*	-0.054
					(0.049)	(0.050)
GDP logged		-0.014		-0.010		-0.045*
		(0.038)		(0.039)		(0.024)
Population logged		-0.026		-0.037		-0.175*
		(0.085)		(0.081)		(0.088)
Trade		-0.000		-0.000		0.001***
		(0.000)		(0.000)		(0.000)
National gini		-0.068		-0.218		0.112
		(0.246)		(0.253)		(0.226)
Share of protestants		-0.046		-0.056		-0.523
		(0.111)		(0.103)		(0.461)
Democracy		0.011		0.006		-0.114**
		(0.015)		(0.015)		(0.044)
Press freedom		0.000		0.000		0.002
		(0.001)		(0.001)		(0.002)
Observations	756	467	756	467	219	128

Standard errors in parentheses

Thresholds: Base. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Political decentralisation

When it comes to the political decentralisation models shown in table 7, the results tell a different story than what is seen with expenditure decentralisation. Firstly, in line with previous decentralisation variables, the benchmarks are insignificant, which might not be as unexpected as was the case with the fiscal decentralisation variables as previous studies have not found as strong of a relationship between political decentralisation and QoG as that between QoG and fiscal decentralisation. The interaction between political decentralisation is marginally significant in the model without controls, but this disappears as controls are introduced. For the ICEDI, however, the interaction in the model without controls is highly significant and negative. While the significance of the interaction drops when controls are introduced, the interaction between political decentralisation and the ICEDI remains negative and statistically significant. This lends weight to H2 as, while grand elite dispersion did not

yield significant results, the inclusion of a unified interest condition did provide a significant effect. Within the model, the coefficient of just political decentralisation is slightly positive while the interaction term is negative. This reflects a dynamic wherein political decentralisation in itself might positively impact QoG, but if such a reform is conducted in a context where elites are dispersed and share unified interests, then political decentralisation leads to worsened QoG. The fact that this interaction is the opposite as to what is seen in fiscal decentralisation is in line with H3, suggesting that political decentralisation in particular is more vulnerable to the detrimental effects of elite capture.

Graph 2 also showcases a more robust marginal effects plot than graph 1, as the confidence intervals remain below zero somewhere after a 0.3 score on the ICEDI. This brings further support to an understanding that political decentralisation has a detrimental effect on QoG the more dispersed and unified in interest grand elites are in a country.

Table 7: Political decentralisation models

Main FE Models: h_f (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
Political decentralisation	-0.028 (0.028)	-0.035 (0.024)	-0.001 (0.016)	-0.022 (0.022)	-0.010 (0.021)	0.010 (0.022)
Political decentralisation##EDI			-0.053* (0.028)	-0.034 (0.040)		
					(.)	(.)
Political decentralisation##ICEDI					-0.114*** (0.026)	-0.244*** (0.062)
EDI			-0.012 (0.027)	0.031 (0.028)		
ICEDI					0.048 (0.031)	0.090** (0.034)
GDP logged		0.033 (0.024)		0.035 (0.024)		-0.032 (0.037)
Population logged		0.052* (0.031)		0.050 (0.031)		0.212** (0.092)
Trade		-0.000 (0.000)		-0.000 (0.000)		0.001* (0.000)
National gini		-0.143 (0.120)		-0.228 (0.146)		-0.090 (0.179)
Share of protestants		-0.025 (0.106)		-0.033 (0.109)		0.093 (0.259)
Democracy		0.001		0.001		0.032

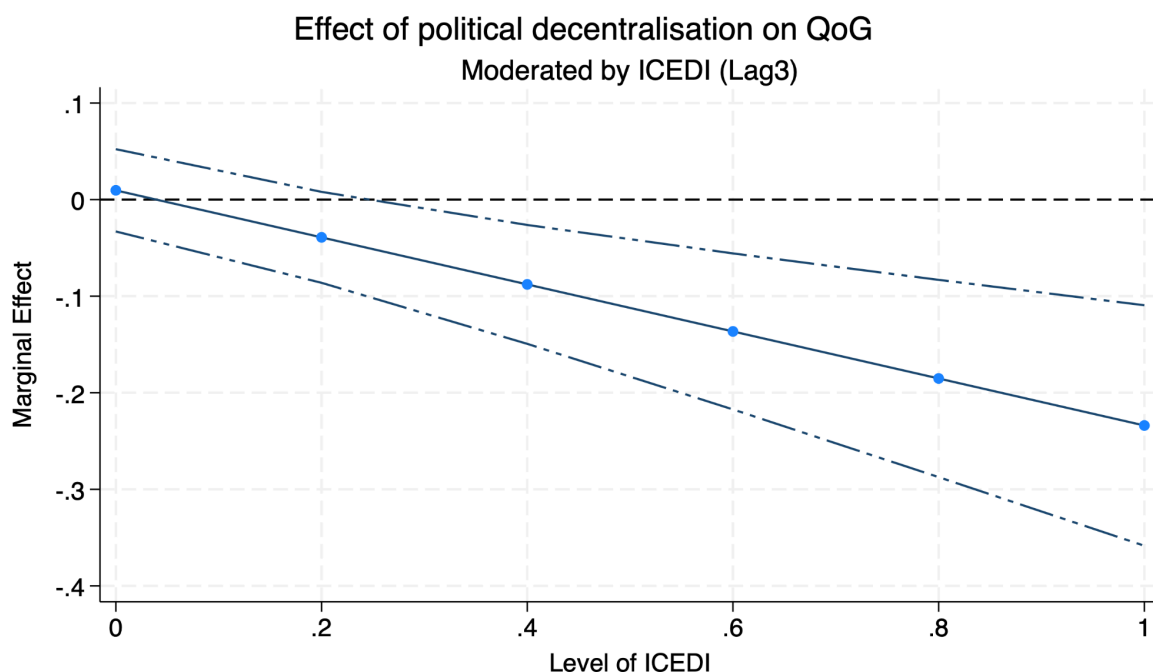
		(0.012)		(0.012)		(0.034)
Press freedom		0.001***		0.001***		0.001*
		(0.000)		(0.000)		(0.001)
Observations	2852	1087	2852	1087	538	207

Standard errors in parentheses

Thresholds: Base. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Graph 2: Interaction between political decentralisation and QoG moderated by ICEDI, FE.



The RAI

In table 8 the benchmark model with just the RAI is once again insignificant, as are the interaction models. Interestingly enough the constituent term for the ICEDI is statistically significant and positive. This suggests that a centralised state will have higher QoG the more dispersed the elites are, although this effect has no effect in either direction as a state decentralises as measured by the RAI.

Table 8: Regional Authority Index models

Main FE Models: n_RAI (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
RAI	0.001	0.001	0.000	0.001	0.003	0.006**
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.003)

RAI##EDI			0.001	-0.001		
			(0.003)	(0.003)		
RAI##ICEDI					-0.002	-0.007
					(0.004)	(0.006)
EDI			0.035	0.067		
			(0.066)	(0.044)		
ICEDI					0.069	0.104*
					(0.054)	(0.057)
GDP logged		0.004		0.009		-0.026
		(0.034)		(0.035)		(0.037)
Population logged		0.063		0.065		0.210*
		(0.063)		(0.062)		(0.110)
Trade		-0.000		-0.000		0.000
		(0.000)		(0.000)		(0.000)
National gini		0.011		-0.099		0.079
		(0.152)		(0.165)		(0.178)
Share of protestants		0.036		0.031		0.162
		(0.140)		(0.138)		(0.250)
Democracy		0.017		0.016		-0.011
		(0.016)		(0.016)		(0.026)
Press freedom		0.001**		0.001**		0.002**
		(0.001)		(0.001)		(0.001)
Observations	1708	734	1708	734	571	259

Standard errors in parentheses

Thresholds: Base. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Robustness checks

The robustness of the results reported above will firstly be tested with alternative time lags, which tests whether the hypothesised causal direction holds under different temporal assumptions as well as whether the effects persist under different time horizons. Further, rigorous robustness tests are especially prudent considering that the moderating variables in this thesis, the EDI and ICEDI, are constructed ad hoc. which is why attention will be paid to test alternative configurations of these so as to approximate their validity.

Due to the large amount of tables generated through the different tests, this section will focus on just the robustness of the results found for expenditure and political decentralisation, which had the most relevant results. Even though these tables will be placed in the appendix, including the tables for all variables would add an excessive amount of pages to an already large appendix¹³.

¹³

Expenditure decentralisation robustness tests

As seen above, expenditure decentralisation has a positive and highly significant effect on QoG when interacted with the ICEDI. If we first look at tables 8-10 which test the same base model across time lags, the results from table 5 are present in table 10 which has a 5 year time lag, but not in tables 8 and 9 which have no time lag and a 1 year lag respectively. This does not necessarily dismiss the robustness of the results, but rather indicate that the effects take a longer time to appear, and that QoG does not change in the immediate aftermath of a fiscal decentralisation reform.

The results are not as consistent in tables 14-17 however, where the threshold for the regional gini coefficient is raised to the 90th percentile (whereas the threshold for the HHI stays at the median). Apart from the expected non-significance of tables 14 and 15 which have no and 1 year time lag respectively, the previously significant results of tables 5 and 10 do not reappear in tables 16 and 17 where variables are lagged 3 and 5 years. This can likely be summed up as the 90th percentile threshold being too stringent to pick up enough cases where inequality is that extreme *and* the economic output is more concentrated for the model to generate significant results, especially given that the number of countries in the data pool for the ICEDI is quite limited as is.

A curious finding is that the EDI generates significant results in tables 14-16 but with a negative coefficient on the interaction with expenditure decentralisation instead of the positive coefficient seen in significant interactions with the ICEDI. This result is seen across time lags of 0, 1 and 3 years. A number of plausible explanations can be behind such a result. For one, it can simply be that the threshold for the EDI used in the main model does not reflect a sufficient power gap between grand elites and the general population for effective local

A guide to the tables in the appendix as they pertain to the different configurations is:

Tables 8-10: Expenditure decentralisation with alternative lags

Tables 11-13: Political decentralisation with alternative lags

Tables 14-17: Expenditure decentralisation with 90th percentile regional gini threshold

Tables 18-21: Political decentralisation with 90th percentile regional gini threshold

Tables 22-25: Expenditure decentralisation with 75th percentile threshold for both regional gini and HHI

Tables 26-29: Political decentralisation with 75th percentile threshold for both regional gini and HHI

Tables 30-33: Expenditure decentralisation with 90th percentile threshold for both regional gini and HHI

Tables 34-37: Political decentralisation with 90th percentile threshold for both regional gini and HHI

Tables 38 and 39: Mexico excluded for expenditure decentralisation and political decentralisation respectively

government capture to take place, which is why the stricter threshold yields significant results where the main versions did not. The fact that the sign of the coefficient changes to negative as opposed to the positive sign of the significant ICEDI interactions, could imply a dynamic where exceptionally powerful but often competing grand elites have a detrimental effect on QoG given fiscal decentralisation, whereas unified elites have a positive one. An alternative explanation could be that the 90th percentile threshold changes the index from measuring countries where elites are dispersed and powerful, to one that rather reflects which countries have a few extremely unequal regions. If that is the case, then the result would suggest that such hubs of regional inequality is rather what impacts QoG negatively when a country decentralises fiscally, rather than elite dispersion as such. Of course, any substantial reasoning that would explain the results to a satisfactory degree would require more rigorous research, and is as such beyond the scope of the thesis.

Moving on to the results of tables 22-25 which raises the threshold of the HHI to the 75th percentile, the same as that of the regional gini for the base model, there is some support for the robustness of the results of table 5. Namely, table 23 which has a 1 year time lag shows a significant and positive interaction under this configuration, which was not seen at the 1 year lag of the base model. This could speak in favour of the robustness of the results, and perhaps imply that the stricter HHI threshold makes for a more precise ICEDI, which captures the effects of the theorised relationship at an earlier stage. The interaction is not significant across any other table with this configuration however, and most importantly it is insignificant in table 24 which uses the same time lag as table 5. This could cast doubts on the robustness of the principal interaction effect found between expenditure decentralisation and QoG.

On the notion of stricter thresholds for the indexes however, tables 30-33 which exhibit the strictest 90th percentile threshold for both regional gini and the HHI show significant and positive interactions in tables 31 and 32 which are lagged 1 and 3 years. Most notably, the coefficient of the ICEDIs interaction in table 32 which has the same 3 year lag as the main table not only retains high significance but shows a much larger effect of 5.094 instead of the 0.195 coefficient in table 5. This could strongly speak in favour of a more rigid index configuration and imply that the positive effect of fiscal decentralisation in the presence of dispersed and unified elites only gets more pronounced the more extreme this factor is.

However, as mentioned in the discussion on the ICEDI construction, Mexico is an outlier in the index and it is plausible that this result merely reflects the development of that country in particular rather than any general results.

Finally, as it concerns table 38 where the outlier Mexico is dropped (threshold configurations remain the same as in the main model), the interaction between expenditure decentralisation and the ICEDI remains significant, although not highly so, in the 3 year lag model and is marginally significant at in the 5 year lag model. This means that the results of the main table, if nothing else, were not driven by the development in Mexico alone, but a trend seen in many other countries.

In sum, the scattered results across the robustness tests cast some doubt over the validity of the results reported in table 5, but the fact that the results that are significant across these tests are in line with what is found in the main model should speak for this finding being fairly robust.

Political decentralisation robustness tests

Turning to political decentralisation, the robustness tests tell a more consistent picture than those for expenditure decentralisation. Firstly, as it pertains to tables 11-13 which test the base model across time lags, the interaction coefficient remains negative and significant across all time lags, and the interaction between political decentralisation and the EDI becomes highly significant and negative in table 13 which has a 5 year time lag. This could suggest that a country merely having dispersed elites could bring with it the same moderating effect as in countries where elites are dispersed and unified in interest, albeit that this effect is delayed.

As for tables 18-21 where the threshold for regional gini is raised to the 90th percentile, the interaction with the ICEDI is significant in table 17 which has the 3 year time lag, but not in tables with other time lags that have this configuration. Unlike what was the case with expenditure decentralisation in tables 14-17, the EDI does not exhibit a significant interaction when regional gini is subject to the stricter 90th percentile threshold.

Where both regional gini and the HHI are under the 75th percentile threshold, in tables 26-29, the interaction between political decentralisation and the ICEDI is significant in tables 27 and

28 which have the 1 and 3 year time lags respectively. While the interaction is insignificant in the other two tables, the fact that it is highly significant in table 28 which has the same 3 year time lag as the main table, should speak for robustness of the results in the main table. This is likewise the case for the strictest 90th percentile thresholds in tables 34-37, where the interaction coefficient for the ICEDI remains significant across tables 34-36, i.e. all time lags except the 5 year time lag. As was the case for expenditure decentralisation and the ICEDI with the strictest threshold configurations, the effect also grows stronger as variables are lagged 3 years. This time going from -0.244 in table 7 to -0.71 in table 36. Finally, table 39 which excludes Mexico showcases significant interactions across all time lags which would indicate that not only do the results hold when this outlier is excluded, but they rather become more robust.

In sum, the interaction between political decentralisation and the ICEDI in particular shows a clear pattern across different time lags and threshold configurations, which is interpreted as these results being robust. With that being said, a limitation that concerns both these results and those in the previous expenditure decentralisation section, is that it should once again be noted that the ICEDI contains quite a small pool of countries. This lessens the generalisability of these results and it could be that the effects identified are as they pertain to the countries in question. However, even though it is limited, the ICEDI does include countries which are not typically seen in other studies with similar limitations. Whereas these studies tend to be limited to just very developed countries, which are also overrepresented in the ICEDI, the index does nonetheless include a number of developing countries. As such the generalisability of the results might not necessarily be as severely limited to just developed countries as has been the case in other studies.

Summary, discussion and further research

In the context of an academic debate where the question about the effects of decentralisation on QoG might seem settled, the results presented in this thesis indicate that this is not the case. None of the decentralisation variables tested show convincing evidence that decentralisation in itself is sufficient for raising QoG. Instead, the thesis has explored the effect of this relationship when state capture by grand elites is made an explicit part of the research design, and hopefully makes a convincing argument for this mechanism's fundamental relevance.

The main contribution of the thesis has been exploring the possibility that the geographic dispersion of grand elites in a country modifies the effects decentralisation reforms have on QoG. This has been tested by utilising a fixed effects panel regression which has indeed produced significant results for the interaction between a measure of elite dispersion and two out of four distinct measures of decentralisation. It is one index in particular, the ICEDI, that is found to significantly interact with expenditure decentralisation and political decentralisation. These results are fairly robust across a variety of tests, as where there are significant results these follow the same dynamic that is seen in the main model tested. The interaction effect does seem to be most robust as it comes to political decentralisation however.

The interaction between the ICEDI and political decentralisation tells of a dynamic wherein the dispersion of grand elites which share interests negatively impact QoG. The implications of this outcome can very well be studied in further detail, but a ready explanation can be found in theories such as that of relative capture outlined by Bardhan and Mookherjee (1999; 2000), a theory which, to the best of my knowledge, has thus far remained untested. The theory outlines requisites for which a local government would become more easily captured by grand elites, which would make for a power disparity between these and the general population. Extrapolating this logic so as to infer what effect this would have on the country level, it is shown that QoG suffers more in countries where grand elites meet these requirements in more regions as compared to countries where they do not meet these requirements to the same degree, given that the country undergoes political decentralisation.

As per H2, the negative effects on QoG as a country decentralises is most prevalent in countries with dispersed elites who also have cohesive interests, this hypothesis is therefore accepted. As it pertains to H1 however, which only concerns dispersion of elites and not their cohesiveness, this hypothesis is not accepted due to a lack of empirical support. It could be that H1 should not be rejected outright however, as the EDI does show some significant interaction with decentralisation across alternative tests, and where found these results do display the same tendencies. As per H3, the results clearly show that political decentralisation is particularly vulnerable to elite capture in comparison to other forms of decentralisation, therefore this hypothesis is accepted. When it comes to how these results fit into the wider academic debate, it could be that they bring some needed clarity to the relationship between political decentralisation and QoG, whose nature has been difficult to determine. As such, the interaction between the ICEDI and political decentralisation points in a clear direction as to in what context specifically political decentralisation will do more harm than good.

Although the interaction with political decentralisation yields the most substantive results, the unexpected results from the interaction with expenditure decentralisation are interesting in that they leave plenty of room for interpretation. Contrary to what is expected in H2 this interaction is positive, meaning that dispersed and unified elites *improve* QoG as fiscal decentralisation is increased rather than hamper it. A more solid explanation for these results might best be left to future research, but the result could fit into the area of fiscal decentralisation research which concerns itself with regional inequality (eg. Kyriacou et al., 2016; Lessmann, 2012; Nguyen et al., 2020). It could be that these results reflect a dynamic where dispersed grand elites in a centralised country have greater incentives to engage in illicit activities, for example bribing officials, in order to avoid redistribution of their wealth to other regions. When a country then decentralises, there is less need for these activities as wealth is kept in its home region to a greater extent. This would lead to relatively higher QoG, at least in the short term, and fits in with the claim that fiscal decentralisation leads to increased QoG. If the explanation suggested above has any truth to it, it introduces an explanation which previous studies lack, namely that corruption is lessened not because decentralisation makes it harder for these practices to take place but rather that there is less need for it. It could of course be the case that fiscal decentralisation lessens corruption and improves QoG some other way, but this is hard to reconcile with the significant interaction

between ICEDI and expenditure decentralisation. If fiscal decentralisation would improve QoG regardless of elite dispersion, not only would the benchmark regression be significant, but the interaction would be insignificant.

To conclude, the results presented in this thesis has brought light to when and how countries are best to decentralise if the aim is to improve QoG. Whereas political decentralisation is shown to have a particularly detrimental effect in countries where elites are dispersed and unified in interest, fiscal decentralisation in the form of expenditure decentralisation is instead beneficial in the same context.

Appendix

Under each section of tables in the appendix the tables for both expenditure and political decentralisation are presented, with the tables covering expenditure decentralisation coming first. The sections are as follows base models with different time lags, 90th percentile regional gini threshold, 75th percentile threshold for both regional gini and HHI, 90th percentile threshold for both regional gini and HHI and finally regressions without Mexico.

Time lag robustness tests

Below are the time lag robustness tests, thresholds are the same as in the main model.

Table 8: Expenditure decentralisation without time lags

Appendix FE Models: expdec (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
(mean) expdec	0.008	-0.002	0.040	0.026	-0.337	-0.583***
	(0.049)	(0.051)	(0.036)	(0.047)	(0.242)	(0.132)
(mean) expdec # (mean) EDI			-0.293*	-0.210		
			(0.158)	(0.128)		
(mean) expdec # (mean) ICEDI					-0.045	-0.038
					(0.100)	(0.120)
revdec						
h_f						
n_RAI						
(mean) EDI			0.135**	0.146**		
			(0.057)	(0.057)		
(mean) ICEDI					0.052	0.060
					(0.055)	(0.058)
ln_gdp		-0.001		0.006		-0.159**
		(0.032)		(0.033)		(0.070)

ln_pop		0.065		0.058		-0.160
		(0.076)		(0.072)		(0.135)
(mean) wdi_trade		-0.001**		-0.001**		-0.000
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.003		-0.162		0.239
		(0.191)		(0.198)		(0.418)
(mean) chprtpt		0.219		0.206		0.592
		(0.179)		(0.180)		(0.370)
(mean) bmr_dem		0.010		0.009		-0.128**
		(0.011)		(0.011)		(0.049)
(mean) press_freedom		0.001*		0.001*		0.004*
		(0.001)		(0.001)		(0.002)
Observations	683	428	683	428	193	112

Standard errors in parentheses

Thresholds: Base. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Expenditure decentralisation with 1 year lag

Appendix FE Models: expdec (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L.(mean) expdec	0.004	-0.041	0.030	-0.001	-0.192	-0.284
	(0.046)	(0.051)	(0.033)	(0.046)	(0.200)	(0.187)
L.(mean) expdec # L.(mean) EDI			-0.232	-0.258*		
			(0.159)	(0.129)		
L.(mean) expdec # L.(mean) ICEDI					0.006	0.074
					(0.085)	(0.097)
revdec						
h_f						
n_RAI						

L.(mean) EDI			0.099	0.135*		
			(0.070)	(0.068)		
L.(mean) ICEDI					0.039	-0.029
					(0.048)	(0.060)
L.ln_gdp		0.006		0.012		-0.057
		(0.031)		(0.032)		(0.089)
L.ln_pop		0.051		0.051		-0.136
		(0.066)		(0.062)		(0.153)
L.(mean) wdi_trade		-0.001**		-0.001**		-0.000
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.028		-0.123		-0.297
		(0.196)		(0.189)		(0.315)
L.(mean) chprtpct		0.089		0.083		0.358
		(0.149)		(0.153)		(0.367)
L.(mean) bmr_dem		0.008		0.011		-0.045
		(0.010)		(0.010)		(0.066)
L.(mean) press_freedom		0.001**		0.001*		0.001
		(0.001)		(0.001)		(0.002)
Observations	685	429	685	429	194	112

Standard errors in parentheses

Thresholds: Base. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Expenditure decentralisation with 5 year lag

Appendix FE Models: expdec (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L5.(mean) expdec	-0.031	-0.087	-0.025	-0.072	-0.080	-0.274***
	(0.052)	(0.093)	(0.053)	(0.101)	(0.078)	(0.075)
L5.(mean) expdec # L5.(mean) EDI			-0.049	-0.083		
			(0.133)	(0.149)		
L5.(mean) expdec # L5.(mean) ICEDI					0.108	0.193**
					(0.064)	(0.069)

revdec						
h_f						
n_RAI						
L5.(mean) EDI			0.018	0.041		
			(0.079)	(0.085)		
L5.(mean) ICEDI					-0.020	-0.071
					(0.044)	(0.041)
L5.ln_gdp		-0.013		-0.011		-0.045
		(0.030)		(0.030)		(0.048)
L5.ln_pop		0.022		0.023		-0.089
		(0.069)		(0.068)		(0.073)
L5.(mean) wdi_trade		-0.000		-0.000		0.001***
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		0.045		0.029		-0.217
		(0.189)		(0.208)		(0.265)
L5.(mean) chprtpct		0.058		0.059		-0.623*
		(0.089)		(0.091)		(0.330)
L5.(mean) bmr_dem		0.017		0.018		-0.104**
		(0.014)		(0.014)		(0.039)
L5.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	622	364	622	364	182	98

Standard errors in parentheses

Thresholds: Base. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Political decentralisation without time lags

Appendix FE Models: h_f (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)

expdec						
revdec						
(mean) h_f=0	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
(mean) h_f=1	0.001	-0.003	0.006	-0.016	0.009	0.047
	(0.016)	(0.013)	(0.023)	(0.025)	(0.025)	(0.028)
(mean) h_f=0 # (mean) EDI			0.000	0.000		
			(.)	(.)		
(mean) h_f=1 # (mean) EDI			-0.004	0.022		
			(0.040)	(0.044)		
(mean) h_f=0 # (mean) ICEDI					0.000	0.000
					(.)	(.)
(mean) h_f=1 # (mean) ICEDI					-0.139**	-0.119**
					(0.060)	(0.054)
n_RAI						
(mean) EDI			-0.027	0.029		
			(0.037)	(0.041)		
(mean) ICEDI					0.028	0.055
					(0.064)	(0.041)
ln_gdp		0.093***		0.095***		0.011
		(0.035)		(0.035)		(0.051)
ln_pop		0.068*		0.066*		0.499***
		(0.035)		(0.035)		(0.150)
(mean) wdi_trade		-0.000		-0.000		0.002**
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.076		-0.163		-0.248
		(0.133)		(0.181)		(0.252)
(mean) chprtpct		-0.002		-0.016		-0.256

		(0.131)		(0.138)		(0.431)
(mean) bmr_dem		-0.002		-0.004		0.040
		(0.015)		(0.016)		(0.063)
(mean) press_freedom		0.000		0.000		0.002
		(0.000)		(0.000)		(0.002)
Observations	2828	1085	2828	1085	535	207

Standard errors in parentheses

Thresholds: Base. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Political decentralisation with 1 year lag

Appendix FE Models: h_f (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L.(mean) h_f	-0.009	-0.019	0.004	-0.028	-0.000	0.052*
	(0.021)	(0.014)	(0.022)	(0.026)	(0.023)	(0.026)
0bL.(mean) h_f # oL.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L.(mean) h_f # L.(mean) EDI			-0.023	0.016		
			(0.037)	(0.048)		
0bL.(mean) h_f # oL.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L.(mean) h_f # L.(mean) ICEDI					-0.134**	-0.127**
					(0.054)	(0.052)
n_RAI						
L.(mean) EDI			-0.028	0.015		

			(0.035)	(0.041)		
L.(mean) ICEDI					0.051	0.049
					(0.056)	(0.039)
L.ln_gdp		0.088***		0.090***		0.028
		(0.029)		(0.030)		(0.047)
L.ln_pop		0.071**		0.070**		0.436***
		(0.035)		(0.035)		(0.152)
L.(mean) wdi_trade		-0.000		-0.000		0.001**
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.049		-0.095		-0.113
		(0.136)		(0.183)		(0.245)
L.(mean) chprtpct		-0.033		-0.041		-0.097
		(0.128)		(0.134)		(0.403)
L.(mean) bmr_dem		-0.002		-0.002		0.012
		(0.014)		(0.014)		(0.045)
L.(mean) press_freedom		0.001		0.001		0.002
		(0.000)		(0.000)		(0.001)
Observations	2836	1085	2836	1085	536	207

Standard errors in parentheses

Thresholds: Base. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Political decentralisation with 5 year lag

Appendix FE Models: h_f (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL5.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L5.(mean) h_f	-0.046	-0.043	-0.003	0.005	-0.027	0.016
	(0.034)	(0.036)	(0.009)	(0.013)	(0.025)	(0.018)

0bL5.(mean) h_f # oL5.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L5.(mean) h_f # L5.(mean) EDI			-0.084***	-0.096***		
			(0.013)	(0.019)		
0bL5.(mean) h_f # oL5.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L5.(mean) h_f # L5.(mean) ICEDI					-0.043*	-0.098**
					(0.022)	(0.046)
n_RAI						
L5.(mean) EDI			-0.010	-0.001		
			(0.020)	(0.026)		
L5.(mean) ICEDI					0.038	0.059
					(0.028)	(0.040)
L5.ln_gdp		0.027		0.025		-0.042
		(0.020)		(0.019)		(0.029)
L5.ln_pop		0.022		0.022		0.114*
		(0.029)		(0.029)		(0.064)
L5.(mean) wdi_trade		-0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		-0.161		-0.144		0.046
		(0.113)		(0.142)		(0.182)
L5.(mean) chprtpct		0.019		0.031		0.227*
		(0.078)		(0.075)		(0.117)
L5.(mean) bmr_dem		-0.001		0.001		0.004
		(0.011)		(0.011)		(0.026)
L5.(mean) press_freedom		0.001***		0.001***		0.001**
		(0.000)		(0.000)		(0.001)
Observations	2869	1094	2869	1094	539	208

Standard errors in parentheses

Thresholds: Base. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

90th percentile, gini

This section contains robustness tests with 90th percentile threshold for regional gini, second quartile for HHI. Time lags of 0,1 3 and 5 years.

Table 14: Expenditure decentralisation with 90th percentile threshold for EDI/ICEDI, no lag

Appendix FE Models: expdec (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
(mean) expdec	0.008	-0.002	0.029	0.024	-0.308	-0.551***
	(0.049)	(0.051)	(0.037)	(0.047)	(0.236)	(0.159)
(mean) expdec # (mean) EDI			-0.342	-0.393**		
			(0.278)	(0.162)		
(mean) expdec # (mean) ICEDI					-1.678*	-0.199
					(0.938)	(1.156)
revdec						
h_f						
n_RAI						
(mean) EDI			0.142	0.196**		
			(0.109)	(0.073)		
(mean) ICEDI					1.004	0.114
					(0.588)	(0.701)
ln_gdp		-0.001		0.003		-0.137
		(0.032)		(0.034)		(0.081)
ln_pop		0.065		0.067		-0.161
		(0.076)		(0.075)		(0.143)
(mean) wdi_trade		-0.001**		-0.001**		-0.001
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.003		-0.046		0.288

		(0.191)		(0.196)		(0.430)
(mean) chprtpt		0.219		0.219		0.539
		(0.179)		(0.179)		(0.401)
(mean) bmr_dem		0.010		0.014		-0.117*
		(0.011)		(0.011)		(0.061)
(mean) press_freedom		0.001*		0.001*		0.003
		(0.001)		(0.001)		(0.002)
Observations	683	428	683	428	193	112

Standard errors in parentheses

Thresholds: Gini90. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 15: Expenditure decentralisation with 90th percentile threshold for EDI/ICEDI, 1 year lag

Appendix FE Models: expdec (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L.(mean) expdec	0.004	-0.041	0.027	-0.004	-0.155	-0.247
	(0.046)	(0.051)	(0.034)	(0.043)	(0.171)	(0.183)
L.(mean) expdec # L.(mean) EDI			-0.336	-0.476***		
			(0.259)	(0.160)		
L.(mean) expdec # L.(mean) ICEDI					-2.141**	-0.618
					(0.933)	(0.784)
revdec						
h_f						
n_RAI						
L.(mean) EDI			0.115	0.202**		
			(0.120)	(0.087)		
L.(mean) ICEDI					1.286**	0.380
					(0.573)	(0.464)

L.ln_gdp		0.006		0.013		-0.069
		(0.031)		(0.033)		(0.085)
L.ln_pop		0.051		0.053		-0.148
		(0.066)		(0.064)		(0.152)
L.(mean) wdi_trade		-0.001**		-0.001**		-0.000
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.028		-0.042		-0.312
		(0.196)		(0.201)		(0.334)
L.(mean) chprtpct		0.089		0.094		0.325
		(0.149)		(0.157)		(0.362)
L.(mean) bmr_dem		0.008		0.015		-0.031
		(0.010)		(0.010)		(0.066)
L.(mean) press_freedom		0.001**		0.001**		0.001
		(0.001)		(0.001)		(0.002)
Observations	685	429	685	429	194	112

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16: Expenditure decentralisation with 90th percentile threshold for EDI/ICEDI, 3 year lag

Appendix FE Models: expdec (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L3.(mean) expdec	-0.005	-0.010	0.016	0.017	-0.068	-0.068
	(0.041)	(0.042)	(0.033)	(0.037)	(0.095)	(0.113)
L3.(mean) expdec # L3.(mean) EDI			-0.316	-0.439**		
			(0.222)	(0.195)		
L3.(mean) expdec # L3.(mean) ICEDI					-0.697	0.017
					(0.575)	(1.216)
revdec						
h_f						

n_RAI						
L3.(mean) EDI			0.113	0.222*		
			(0.131)	(0.115)		
L3.(mean) ICEDI					0.429	-0.008
					(0.359)	(0.762)
L3.ln_gdp		-0.003		0.002		0.001
		(0.036)		(0.037)		(0.050)
L3.ln_pop		0.010		0.012		-0.102
		(0.092)		(0.091)		(0.121)
L3.(mean) wdi_trade		-0.001**		-0.001**		0.001
		(0.000)		(0.000)		(0.001)
L3.(mean) natgini_		-0.030		-0.081		0.062
		(0.257)		(0.269)		(0.179)
L3.(mean) chprtpt		0.016		0.015		-0.385
		(0.113)		(0.116)		(0.411)
L3.(mean) bmr_dem		0.024**		0.029**		-0.088
		(0.012)		(0.012)		(0.054)
L3.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	689	431	689	431	196	112

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 17: Expenditure decentralisation with 90th percentile threshold for EDI/ICEDI, 5 year lag

Appendix FE Models: expdec (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L5.(mean) expdec	-0.031	-0.087	-0.024	-0.075	-0.076	-0.228**
	(0.052)	(0.093)	(0.052)	(0.095)	(0.071)	(0.101)

L5.(mean) expdec # L5.(mean) EDI			-0.096	-0.216		
			(0.236)	(0.232)		
L5.(mean) expdec # L5.(mean) ICEDI					-1.097*	-1.050
					(0.535)	(0.609)
revdec						
h_f						
n_RAI						
L5.(mean) EDI			0.034	0.120		
			(0.143)	(0.135)		
L5.(mean) ICEDI					0.728**	0.704*
					(0.321)	(0.366)
L5.ln_gdp		-0.013		-0.010		-0.059
		(0.030)		(0.031)		(0.045)
L5.ln_pop		0.022		0.027		-0.116
		(0.069)		(0.070)		(0.082)
L5.(mean) wdi_trade		-0.000		-0.000		0.001***
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		0.045		0.017		-0.280
		(0.189)		(0.200)		(0.277)
L5.(mean) chprtpt		0.058		0.057		-0.581*
		(0.089)		(0.090)		(0.291)
L5.(mean) bmr_dem		0.017		0.020		-0.078*
		(0.014)		(0.015)		(0.043)
L5.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	622	364	622	364	182	98

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 18: Political decentralisation with 90th percentile threshold for EDI/ICEDI, no lag

Appendix FE Models: h_f (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
(mean) h_f=0	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
(mean) h_f=1	0.001	-0.003	-0.028	-0.037	-0.009	0.044
	(0.016)	(0.013)	(0.049)	(0.042)	(0.018)	(0.029)
(mean) h_f=0 # (mean) EDI			0.000	0.000		
			(.)	(.)		
(mean) h_f=1 # (mean) EDI			0.102	0.111		
			(0.086)	(0.074)		
(mean) h_f=0 # (mean) ICEDI					0.000	0.000
					(.)	(.)
(mean) h_f=1 # (mean) ICEDI					-0.117*	-0.126*
					(0.062)	(0.067)
n_RAI						
(mean) EDI			-0.051	-0.053		
			(0.032)	(0.034)		
(mean) ICEDI					-0.009	0.110
					(0.066)	(0.069)
ln_gdp		0.093***		0.091***		0.019
		(0.035)		(0.035)		(0.054)
ln_pop		0.068*		0.074**		0.520***
		(0.035)		(0.035)		(0.140)

(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.076		-0.001		-0.285
		(0.133)		(0.144)		(0.268)
(mean) chprtpct		-0.002		0.000		-0.277
		(0.131)		(0.126)		(0.445)
(mean) bmr_dem		-0.002		-0.003		0.035
		(0.015)		(0.015)		(0.062)
(mean) press_freedom		0.000		0.000		0.002
		(0.000)		(0.000)		(0.002)
Observations	2828	1085	2828	1085	535	207

Standard errors in parentheses

Thresholds: Gini90. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 19: Political decentralisation with 90th percentile threshold for EDI/ICEDI, 1 year lag

Appendix FE Models: h_f (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L.(mean) h_f	-0.009	-0.019	-0.028	-0.052	-0.026	0.052*
	(0.021)	(0.014)	(0.046)	(0.046)	(0.026)	(0.027)
0bL.(mean) h_f # oL.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L.(mean) h_f # L.(mean) EDI			0.075	0.118		
			(0.082)	(0.085)		
0bL.(mean) h_f # oL.(mean) ICEDI					0.000	0.000
					(.)	(.)

1L.(mean) h_f # L.(mean) ICEDI					-0.089	-0.090
					(0.064)	(0.070)
n_RAI						
L.(mean) EDI			-0.052*	-0.071**		
			(0.031)	(0.035)		
L.(mean) ICEDI					-0.013	0.065
					(0.076)	(0.063)
L.ln_gdp		0.088***		0.085***		0.034
		(0.029)		(0.029)		(0.050)
L.ln_pop		0.071**		0.081**		0.456***
		(0.035)		(0.035)		(0.145)
L.(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.049		0.054		-0.142
		(0.136)		(0.143)		(0.258)
L.(mean) chprtpct		-0.033		-0.031		-0.136
		(0.128)		(0.124)		(0.427)
L.(mean) bmr_dem		-0.002		-0.002		0.008
		(0.014)		(0.014)		(0.044)
L.(mean) press_freedom		0.001		0.001		0.002
		(0.000)		(0.000)		(0.002)
Observations	2836	1085	2836	1085	536	207

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 20: Political decentralisation with 90th percentile threshold for EDI/ICEDI, 3 year lag

Appendix FE Models: h_f (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						

revdec						
0bL3.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L3.(mean) h_f	-0.028	-0.035	-0.026	-0.043	-0.019	0.011
	(0.028)	(0.024)	(0.036)	(0.039)	(0.027)	(0.022)
0bL3.(mean) h_f # oL3.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L3.(mean) h_f # L3.(mean) EDI			0.011	0.031		
			(0.063)	(0.072)		
0bL3.(mean) h_f # oL3.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L3.(mean) h_f # L3.(mean) ICEDI					-0.126**	-0.227**
					(0.059)	(0.103)
n_RAI						
L3.(mean) EDI			-0.036	-0.028		
			(0.027)	(0.033)		
L3.(mean) ICEDI					0.001	0.079
					(0.067)	(0.056)
L3.ln_gdp		0.033		0.032		-0.021
		(0.024)		(0.023)		(0.043)
L3.ln_pop		0.052*		0.056*		0.246**
		(0.031)		(0.032)		(0.105)
L3.(mean) wdi_trade		-0.000		-0.000		0.001*
		(0.000)		(0.000)		(0.000)
L3.(mean) natgini_		-0.143		-0.101		-0.130
		(0.120)		(0.134)		(0.210)
L3.(mean) chprtpct		-0.025		-0.025		0.008
		(0.106)		(0.107)		(0.326)
L3.(mean) bmr_dem		0.001		0.001		0.025

		(0.012)		(0.012)		(0.035)
L3.(mean) press_freedom		0.001***		0.001***		0.002
		(0.000)		(0.000)		(0.001)
Observations	2852	1087	2852	1087	538	207

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 21: Political decentralisation with 90th percentile threshold for EDI/ICEDI, 5 year lag

Appendix FE Models: h_f (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL5.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L5.(mean) h_f	-0.046	-0.043	-0.028	-0.022	-0.028	0.016
	(0.034)	(0.036)	(0.030)	(0.031)	(0.024)	(0.018)
0bL5.(mean) h_f # oL5.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L5.(mean) h_f # L5.(mean) EDI			-0.033	-0.043		
			(0.049)	(0.051)		
0bL5.(mean) h_f # oL5.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L5.(mean) h_f # L5.(mean) ICEDI					-0.071	-0.101
					(0.042)	(0.083)
n_RAI						
L5.(mean) EDI			-0.030	-0.032		
			(0.023)	(0.035)		
L5.(mean) ICEDI					0.023	0.040

					(0.047)	(0.056)
L5.ln_gdp		0.027		0.025		-0.037
		(0.020)		(0.019)		(0.032)
L5.ln_pop		0.022		0.027		0.133**
		(0.029)		(0.029)		(0.065)
L5.(mean) wdi_trade		-0.000		-0.000		0.000*
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		-0.161		-0.104		0.034
		(0.113)		(0.130)		(0.199)
L5.(mean) chprtpt		0.019		0.016		0.200*
		(0.078)		(0.082)		(0.118)
L5.(mean) bmr_dem		-0.001		0.001		-0.000
		(0.011)		(0.011)		(0.027)
L5.(mean) press_freedom		0.001***		0.001***		0.002**
		(0.000)		(0.000)		(0.001)
Observations	2869	1094	2869	1094	539	208

Standard errors in parentheses

Thresholds: Gini90. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

75th percentile, gini and HHI

This section contains the robustness tests where the threshold for HHI is raised to the 75th percentile, the same as the regional gini threshold for the main model. The regional gini threshold also sits at the third quartile.

Table 22: Expenditure decentralisation, Q3 threshold for both regional gini and HHI, no time lag

Appendix FE Models: expdec (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
(mean) expdec	0.008	-0.002	0.040	0.026	-0.347	-0.616***
	(0.049)	(0.051)	(0.036)	(0.047)	(0.248)	(0.142)
(mean) expdec # (mean) EDI			-0.293*	-0.210		
			(0.158)	(0.128)		

(mean) expdec # (mean)					0.391**	0.253
ICEDI					(0.181)	(0.348)
revdec						
h_f						
n_RAI						
(mean) EDI			0.135**	0.146**		
			(0.057)	(0.057)		
(mean) ICEDI					-0.094	-0.043
					(0.086)	(0.188)
ln_gdp		-0.001		0.006		-0.151**
		(0.032)		(0.033)		(0.060)
ln_pop		0.065		0.058		-0.161
		(0.076)		(0.072)		(0.136)
(mean) wdi_trade		-0.001**		-0.001**		-0.001
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.003		-0.162		0.294
		(0.191)		(0.198)		(0.406)
(mean) chprtptct		0.219		0.206		0.539
		(0.179)		(0.180)		(0.385)
(mean) bmr_dem		0.010		0.009		-0.134**
		(0.011)		(0.011)		(0.051)
(mean) press_freedom		0.001*		0.001*		0.004*
		(0.001)		(0.001)		(0.002)
Observations	683	428	683	428	193	112

Standard errors in parentheses

Thresholds: HHI75. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 23: Expenditure decentralisation, Q3 threshold for both regional gini and HHI, 1 year time lag

Appendix FE Models: expdec (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L.(mean) expdec	0.004	-0.041	0.030	-0.001	-0.198	-0.309*
	(0.046)	(0.051)	(0.033)	(0.046)	(0.205)	(0.179)
L.(mean) expdec # L.(mean) EDI			-0.232	-0.258*		
			(0.159)	(0.129)		
L.(mean) expdec # L.(mean) ICEDI					0.471***	0.722**
					(0.141)	(0.333)
revdec						
h_f						
n_RAI						
L.(mean) EDI			0.099	0.135*		
			(0.070)	(0.068)		
L.(mean) ICEDI					-0.133*	-0.353*
					(0.072)	(0.200)
L.ln_gdp		0.006		0.012		-0.057
		(0.031)		(0.032)		(0.080)
L.ln_pop		0.051		0.051		-0.138
		(0.066)		(0.062)		(0.151)
L.(mean) wdi_trade		-0.001**		-0.001**		-0.000
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.028		-0.123		-0.260
		(0.196)		(0.189)		(0.327)
L.(mean) chprtpct		0.089		0.083		0.325
		(0.149)		(0.153)		(0.352)
L.(mean) bmr_dem		0.008		0.011		-0.050
		(0.010)		(0.010)		(0.062)

L.(mean) press_freedom		0.001**		0.001*		0.001
		(0.001)		(0.001)		(0.002)
Observations	685	429	685	429	194	112

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 24: Expenditure decentralisation, Q3 threshold for both regional gini and HHI, 3 year time lag

Appendix FE Models: expdec (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L3.(mean) expdec	-0.005	-0.010	0.012	0.003	-0.080	-0.097
	(0.041)	(0.042)	(0.036)	(0.049)	(0.103)	(0.101)
L3.(mean) expdec # L3.(mean) EDI			-0.154	-0.135		
			(0.138)	(0.139)		
L3.(mean) expdec # L3.(mean) ICEDI					0.404*	0.218
					(0.214)	(0.645)
revdec						
h_f						
n_RAI						
L3.(mean) EDI			0.076	0.122		
			(0.079)	(0.083)		
L3.(mean) ICEDI					-0.137	-0.068
					(0.119)	(0.394)
L3.ln_gdp		-0.003		0.003		-0.008
		(0.036)		(0.036)		(0.058)
L3.ln_pop		0.010		0.000		-0.105
		(0.092)		(0.086)		(0.120)
L3.(mean) wdi_trade		-0.001**		-0.001**		0.001

		(0.000)		(0.000)		(0.001)
L3.(mean) natgini_		-0.030		-0.197		0.074
		(0.257)		(0.260)		(0.173)
L3.(mean) chprtpct		0.016		0.000		-0.403
		(0.113)		(0.110)		(0.416)
L3.(mean) bmr_dem		0.024**		0.021*		-0.095**
		(0.012)		(0.012)		(0.045)
L3.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	689	431	689	431	196	112

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 25: Expenditure decentralisation, Q3 threshold for both regional gini and HHI, 5 year time lag

Appendix FE Models: expdec (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L5.(mean) expdec	-0.031	-0.087	-0.025	-0.072	-0.068	-0.276***
	(0.052)	(0.093)	(0.053)	(0.101)	(0.078)	(0.085)
L5.(mean) expdec # L5.(mean) EDI			-0.049	-0.083		
			(0.133)	(0.149)		
L5.(mean) expdec # L5.(mean) ICEDI					0.433**	0.506*
					(0.160)	(0.261)
revdec						
h_f						
n_RAI						
L5.(mean) EDI			0.018	0.041		
			(0.079)	(0.085)		

L5.(mean) ICEDI					-0.213***	-0.238
					(0.066)	(0.138)
L5.ln_gdp		-0.013		-0.011		-0.050
		(0.030)		(0.030)		(0.043)
L5.ln_pop		0.022		0.023		-0.094
		(0.069)		(0.068)		(0.076)
L5.(mean) wdi_trade		-0.000		-0.000		0.001***
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		0.045		0.029		-0.178
		(0.189)		(0.208)		(0.274)
L5.(mean) chprtpct		0.058		0.059		-0.656*
		(0.089)		(0.091)		(0.353)
L5.(mean) bmr_dem		0.017		0.018		-0.108**
		(0.014)		(0.014)		(0.039)
L5.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	622	364	622	364	182	98

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 26: Political decentralisation, Q3 threshold for both regional gini and HHI, no time lag

Appendix FE Models: h_f (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
(mean) h_f=0	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
(mean) h_f=1	0.001	-0.003	0.006	-0.016	-0.017	0.049*
	(0.016)	(0.013)	(0.023)	(0.025)	(0.020)	(0.028)

(mean) h_f=0 # (mean) EDI			0.000	0.000		
			(.)	(.)		
(mean) h_f=1 # (mean) EDI			-0.004	0.022		
			(0.040)	(0.044)		
(mean) h_f=0 # (mean) ICEDI					0.000	0.000
					(.)	(.)
(mean) h_f=1 # (mean) ICEDI					-0.176***	-0.195*
					(0.065)	(0.102)
n_RAI						
(mean) EDI			-0.027	0.029		
			(0.037)	(0.041)		
(mean) ICEDI					0.056	0.071
					(0.055)	(0.108)
ln_gdp		0.093***		0.095***		0.012
		(0.035)		(0.035)		(0.050)
ln_pop		0.068*		0.066*		0.517***
		(0.035)		(0.035)		(0.147)
(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.076		-0.163		-0.226
		(0.133)		(0.181)		(0.243)
(mean) chprtptct		-0.002		-0.016		-0.271
		(0.131)		(0.138)		(0.428)
(mean) bmr_dem		-0.002		-0.004		0.040
		(0.015)		(0.016)		(0.063)
(mean) press_freedom		0.000		0.000		0.002
		(0.000)		(0.000)		(0.002)
Observations	2828	1085	2828	1085	535	207

Standard errors in parentheses

Thresholds: HHI75. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 27: Political decentralisation, Q3 threshold for both regional gini and HHI, 1 year time lag

Appendix FE Models: h_f (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L.(mean) h_f	-0.009	-0.019	0.004	-0.028	-0.015	0.054**
	(0.021)	(0.014)	(0.022)	(0.026)	(0.018)	(0.026)
0bL.(mean) h_f # oL.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L.(mean) h_f # L.(mean) EDI			-0.023	0.016		
			(0.037)	(0.048)		
0bL.(mean) h_f # oL.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L.(mean) h_f # L.(mean) ICEDI					-0.203***	-0.213**
					(0.069)	(0.098)
n_RAI						
L.(mean) EDI			-0.028	0.015		
			(0.035)	(0.041)		
L.(mean) ICEDI					0.121**	0.072
					(0.052)	(0.104)
L.ln_gdp		0.088***		0.090***		0.028
		(0.029)		(0.030)		(0.046)
L.ln_pop		0.071**		0.070**		0.452***

		(0.035)		(0.035)		(0.149)
L.(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.049		-0.095		-0.093
		(0.136)		(0.183)		(0.240)
L.(mean) chprtpct		-0.033		-0.041		-0.112
		(0.128)		(0.134)		(0.404)
L.(mean) bmr_dem		-0.002		-0.002		0.012
		(0.014)		(0.014)		(0.045)
L.(mean) press_freedom		0.001		0.001		0.002
		(0.000)		(0.000)		(0.001)
Observations	2836	1085	2836	1085	536	207

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 28: Political decentralisation, Q3 threshold for both regional gini and HHI, 3 year time lag

Appendix FE Models: h_f (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL3.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L3.(mean) h_f	-0.028	-0.035	-0.001	-0.022	-0.024	0.013
	(0.028)	(0.024)	(0.016)	(0.022)	(0.026)	(0.023)
0bL3.(mean) h_f # oL3.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L3.(mean) h_f # L3.(mean) EDI			-0.053*	-0.034		
			(0.028)	(0.040)		

0bL3.(mean) h_f # oL3.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L3.(mean) h_f # L3.(mean) ICEDI					-0.157***	-0.373***
					(0.040)	(0.125)
n_RAI						
L3.(mean) EDI			-0.012	0.031		
			(0.027)	(0.028)		
L3.(mean) ICEDI					0.109**	0.160
					(0.042)	(0.120)
L3.ln_gdp		0.033		0.035		-0.030
		(0.024)		(0.024)		(0.036)
L3.ln_pop		0.052*		0.050		0.244**
		(0.031)		(0.031)		(0.104)
L3.(mean) wdi_trade		-0.000		-0.000		0.001*
		(0.000)		(0.000)		(0.000)
L3.(mean) natgini_		-0.143		-0.228		-0.052
		(0.120)		(0.146)		(0.189)
L3.(mean) chprtpt		-0.025		-0.033		0.041
		(0.106)		(0.109)		(0.266)
L3.(mean) bmr_dem		0.001		0.001		0.032
		(0.012)		(0.012)		(0.035)
L3.(mean) press_freedom		0.001***		0.001***		0.001
		(0.000)		(0.000)		(0.001)
Observations	2852	1087	2852	1087	538	207

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 29: Political decentralisation, Q3 threshold for both regional gini and HHI, 5 year time lag

Appendix FE Models: h_f (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
--	-----	-----	-----	-----	-----	-----

	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL5.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L5.(mean) h_f	-0.046	-0.043	-0.003	0.005	-0.038	0.017
	(0.034)	(0.036)	(0.009)	(0.013)	(0.031)	(0.018)
0bL5.(mean) h_f # oL5.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L5.(mean) h_f # L5.(mean) EDI			-0.084***	-0.096***		
			(0.013)	(0.019)		
0bL5.(mean) h_f # oL5.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L5.(mean) h_f # L5.(mean) ICEDI					-0.034	-0.157
					(0.034)	(0.094)
n_RAI						
L5.(mean) EDI			-0.010	-0.001		
			(0.020)	(0.026)		
L5.(mean) ICEDI					0.035	0.076
					(0.039)	(0.085)
L5.ln_gdp		0.027		0.025		-0.041
		(0.020)		(0.019)		(0.029)
L5.ln_pop		0.022		0.022		0.131*
		(0.029)		(0.029)		(0.067)
L5.(mean) wdi_trade		-0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		-0.161		-0.144		0.068
		(0.113)		(0.142)		(0.195)

L5.(mean) chprtpct		0.019		0.031		0.207*
		(0.078)		(0.075)		(0.108)
L5.(mean) bmr_dem		-0.001		0.001		0.003
		(0.011)		(0.011)		(0.027)
L5.(mean) press_freedom		0.001***		0.001***		0.001*
		(0.000)		(0.000)		(0.001)
Observations	2869	1094	2869	1094	539	208

Standard errors in parentheses

Thresholds: HHI75. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

90th percentile, gini and HHI

This section includes the robustness tests with the strictest thresholds. Both regional gini and HHI have a threshold of the 90th percentile

Table 30: Expenditure decentralisation, 90th percentile threshold for both regional gini and HHI, no time lag

Appendix FE Models: expdec (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
(mean) expdec	0.008	-0.002	0.029	0.024	-0.342	-0.589***
	(0.049)	(0.051)	(0.037)	(0.047)	(0.244)	(0.135)
(mean) expdec # (mean) EDI			-0.342	-0.393**		
			(0.278)	(0.162)		
(mean) expdec # (mean) ICEDI					0.138	1.030
					(2.475)	(1.159)
revdec						
h_f						
n_RAI						
(mean) EDI			0.142	0.196**		
			(0.109)	(0.073)		

(mean) ICEDI					0.387	-0.322
					(1.533)	(0.511)
ln_gdp		-0.001		0.003		-0.145**
		(0.032)		(0.034)		(0.065)
ln_pop		0.065		0.067		-0.164
		(0.076)		(0.075)		(0.131)
(mean) wdi_trade		-0.001**		-0.001**		-0.001
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.003		-0.046		0.274
		(0.191)		(0.196)		(0.418)
(mean) chprtpct		0.219		0.219		0.579
		(0.179)		(0.179)		(0.381)
(mean) bmr_dem		0.010		0.014		-0.128**
		(0.011)		(0.011)		(0.048)
(mean) press_freedom		0.001*		0.001*		0.004*
		(0.001)		(0.001)		(0.002)
Observations	683	428	683	428	193	112

Standard errors in parentheses

Thresholds: Strict90. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 31: Expenditure decentralisation, 90th percentile threshold for both regional gini and HHI, 1 year time lag

Appendix FE Models: expdec (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L.(mean) expdec	0.004	-0.041	0.027	-0.004	-0.195	-0.302*
	(0.046)	(0.051)	(0.034)	(0.043)	(0.201)	(0.174)
L.(mean) expdec # L.(mean) EDI			-0.336	-0.476***		
			(0.259)	(0.160)		
L.(mean) expdec # L.(mean) ICEDI					0.576	1.413**
					(2.333)	(0.660)
revdec						

h_f						
n_RAI						
L.(mean) EDI			0.115	0.202**		
			(0.120)	(0.087)		
L.(mean) ICEDI					0.198	-0.545*
					(1.439)	(0.305)
L.ln_gdp		0.006		0.013		-0.073
		(0.031)		(0.033)		(0.075)
L.ln_pop		0.051		0.053		-0.146
		(0.066)		(0.064)		(0.148)
L.(mean) wdi_trade		-0.001**		-0.001**		-0.000
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.028		-0.042		-0.317
		(0.196)		(0.201)		(0.335)
L.(mean) chprtpct		0.089		0.094		0.360
		(0.149)		(0.157)		(0.376)
L.(mean) bmr_dem		0.008		0.015		-0.051
		(0.010)		(0.010)		(0.062)
L.(mean) press_freedom		0.001**		0.001**		0.001
		(0.001)		(0.001)		(0.002)
Observations	685	429	685	429	194	112

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 32: Expenditure decentralisation, 90th percentile threshold for both regional gini and HHI, 3 year time lag

Appendix FE Models: expdec (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)

L3.(mean) expdec	-0.005	-0.010	0.016	0.017	-0.083	-0.084
	(0.041)	(0.042)	(0.033)	(0.037)	(0.103)	(0.097)
L3.(mean) expdec # L3.(mean) EDI			-0.316	-0.439**		
			(0.222)	(0.195)		
L3.(mean) expdec # L3.(mean) ICEDI					4.212*	5.094***
					(2.033)	(1.254)
revdec						
h_f						
n_RAI						
L3.(mean) EDI			0.113	0.222*		
			(0.131)	(0.115)		
L3.(mean) ICEDI					-2.281*	-2.953***
					(1.211)	(0.641)
L3.ln_gdp		-0.003		0.002		0.002
		(0.036)		(0.037)		(0.049)
L3.ln_pop		0.010		0.012		-0.103
		(0.092)		(0.091)		(0.120)
L3.(mean) wdi_trade		-0.001**		-0.001**		0.001
		(0.000)		(0.000)		(0.001)
L3.(mean) natgini_		-0.030		-0.081		0.063
		(0.257)		(0.269)		(0.165)
L3.(mean) chprtpct		0.016		0.015		-0.375
		(0.113)		(0.116)		(0.402)
L3.(mean) bmr_dem		0.024**		0.029**		-0.091*
		(0.012)		(0.012)		(0.044)
L3.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	689	431	689	431	196	112

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 33: Expenditure decentralisation, 90th percentile threshold for both regional gini and HHI, 5 year time lag

Appendix FE Models: expdec (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
L5.(mean) expdec	-0.031	-0.087	-0.024	-0.075	-0.078	-0.270***
	(0.052)	(0.093)	(0.052)	(0.095)	(0.080)	(0.089)
L5.(mean) expdec # L5.(mean) EDI			-0.096	-0.216		
			(0.236)	(0.232)		
L5.(mean) expdec # L5.(mean) ICEDI					-0.686	-0.228
					(0.832)	(0.481)
revdec						
h_f						
n_RAI						
L5.(mean) EDI			0.034	0.120		
			(0.143)	(0.135)		
L5.(mean) ICEDI					0.716	0.363*
					(0.496)	(0.201)
L5.ln_gdp		-0.013		-0.010		-0.061
		(0.030)		(0.031)		(0.044)
L5.ln_pop		0.022		0.027		-0.105
		(0.069)		(0.070)		(0.084)
L5.(mean) wdi_trade		-0.000		-0.000		0.001***
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		0.045		0.017		-0.222
		(0.189)		(0.200)		(0.272)

L5.(mean) chprtpct		0.058		0.057		-0.649*
		(0.089)		(0.090)		(0.357)
L5.(mean) bmr_dem		0.017		0.020		-0.109**
		(0.014)		(0.015)		(0.040)
L5.(mean) press_freedom		-0.000		-0.000		0.002
		(0.001)		(0.001)		(0.001)
Observations	622	364	622	364	182	98

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 34: Political decentralisation, 90th percentile threshold for both regional gini and HHI, no time lag

Appendix FE Models: h_f (Current)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
(mean) h_f=0	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
(mean) h_f=1	0.001	-0.003	-0.028	-0.037	-0.029	0.051*
	(0.016)	(0.013)	(0.049)	(0.042)	(0.028)	(0.027)
(mean) h_f=0 # (mean) EDI			0.000	0.000		
			(.)	(.)		
(mean) h_f=1 # (mean) EDI			0.102	0.111		
			(0.086)	(0.074)		
(mean) h_f=0 # (mean) ICEDI					0.000	0.000
					(.)	(.)
(mean) h_f=1 # (mean) ICEDI					-0.466	-0.369***
					(0.323)	(0.121)
n_RAI						

(mean) EDI			-0.051	-0.053		
			(0.032)	(0.034)		
(mean) ICEDI					0.052	0.162
					(0.357)	(0.131)
ln_gdp		0.093***		0.091***		0.016
		(0.035)		(0.035)		(0.052)
ln_pop		0.068*		0.074**		0.519***
		(0.035)		(0.035)		(0.145)
(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
(mean) natgini_		-0.076		-0.001		-0.255
		(0.133)		(0.144)		(0.260)
(mean) chprtptct		-0.002		0.000		-0.292
		(0.131)		(0.126)		(0.438)
(mean) bmr_dem		-0.002		-0.003		0.037
		(0.015)		(0.015)		(0.063)
(mean) press_freedom		0.000		0.000		0.002
		(0.000)		(0.000)		(0.002)
Observations	2828	1085	2828	1085	535	207

Standard errors in parentheses

Thresholds: Strict90. Year FE included. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 35: Political decentralisation, 90th percentile threshold for both regional gini and HHI, 1 year time lag

Appendix FE Models: h_f (Lag1)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						

0bL.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L.(mean) h_f	-0.009	-0.019	-0.028	-0.052	-0.040	0.056**
	(0.021)	(0.014)	(0.046)	(0.046)	(0.032)	(0.026)
0bL.(mean) h_f # oL.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L.(mean) h_f # L.(mean) EDI			0.075	0.118		
			(0.082)	(0.085)		
0bL.(mean) h_f # oL.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L.(mean) h_f # L.(mean) ICEDI					-0.399	-0.279**
					(0.311)	(0.117)
n_RAI						
L.(mean) EDI			-0.052*	-0.071**		
			(0.031)	(0.035)		
L.(mean) ICEDI					0.094	0.042
					(0.344)	(0.085)
L.ln_gdp		0.088***		0.085***		0.032
		(0.029)		(0.029)		(0.049)
L.ln_pop		0.071**		0.081**		0.454***
		(0.035)		(0.035)		(0.148)
L.(mean) wdi_trade		-0.000		-0.000		0.002***
		(0.000)		(0.000)		(0.001)
L.(mean) natgini_		-0.049		0.054		-0.121
		(0.136)		(0.143)		(0.253)
L.(mean) chprtpct		-0.033		-0.031		-0.139
		(0.128)		(0.124)		(0.416)
L.(mean) bmr_dem		-0.002		-0.002		0.009
		(0.014)		(0.014)		(0.045)
L.(mean) press_freedom		0.001		0.001		0.002

		(0.000)		(0.000)		(0.002)
Observations	2836	1085	2836	1085	536	207

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L1. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 36: Political decentralisation, 90th percentile threshold for both regional gini and HHI, 3 year time lag

Appendix FE Models: h_f (Lag3)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL3.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L3.(mean) h_f	-0.028	-0.035	-0.026	-0.043	-0.040	0.016
	(0.028)	(0.024)	(0.036)	(0.039)	(0.039)	(0.022)
0bL3.(mean) h_f # oL3.(mean) EDI			0.000	0.000		
			(.)	(.)		
1L3.(mean) h_f # L3.(mean) EDI			0.011	0.031		
			(0.063)	(0.072)		
0bL3.(mean) h_f # oL3.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L3.(mean) h_f # L3.(mean) ICEDI					-0.475	-0.781***
					(0.355)	(0.141)
n_RAI						
L3.(mean) EDI			-0.036	-0.028		
			(0.027)	(0.033)		
L3.(mean) ICEDI					0.126	0.110
					(0.376)	(0.097)

L3.ln_gdp		0.033		0.032		-0.025
		(0.024)		(0.023)		(0.041)
L3.ln_pop		0.052*		0.056*		0.243**
		(0.031)		(0.032)		(0.107)
L3.(mean) wdi_trade		-0.000		-0.000		0.001*
		(0.000)		(0.000)		(0.000)
L3.(mean) natgini_		-0.143		-0.101		-0.101
		(0.120)		(0.134)		(0.204)
L3.(mean) chprtpt		-0.025		-0.025		0.024
		(0.106)		(0.107)		(0.294)
L3.(mean) bmr_dem		0.001		0.001		0.027
		(0.012)		(0.012)		(0.036)
L3.(mean) press_freedom		0.001***		0.001***		0.002
		(0.000)		(0.000)		(0.001)
Observations	2852	1087	2852	1087	538	207

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L3. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 37: Political decentralisation, 90th percentile threshold for both regional gini and HHI, 5 year time lag

Appendix FE Models: h_f (Lag5)

	(1)	(2)	(3)	(4)	(5)	(6)
	Bench (NoCtrl)	Bench (Ctrl)	EDI (NoCtrl)	EDI (Ctrl)	ICEDI (NoCtrl)	ICEDI (Ctrl)
expdec						
revdec						
0bL5.(mean) h_f	0.000	0.000	0.000	0.000	0.000	0.000
	(.)	(.)	(.)	(.)	(.)	(.)
1L5.(mean) h_f	-0.046	-0.043	-0.028	-0.022	-0.030	0.018
	(0.034)	(0.036)	(0.030)	(0.031)	(0.024)	(0.018)
0bL5.(mean) h_f # oL5.(mean) EDI			0.000	0.000		

			(.)	(.)		
1L5.(mean) h_f # L5.(mean) EDI			-0.033	-0.043		
			(0.049)	(0.051)		
0bL5.(mean) h_f # oL5.(mean) ICEDI					0.000	0.000
					(.)	(.)
1L5.(mean) h_f # L5.(mean) ICEDI					-0.461*	-0.412
					(0.270)	(0.262)
n_RAI						
L5.(mean) EDI			-0.030	-0.032		
			(0.023)	(0.035)		
L5.(mean) ICEDI					0.418	0.177
					(0.285)	(0.243)
L5.ln_gdp		0.027		0.025		-0.038
		(0.020)		(0.019)		(0.030)
L5.ln_pop		0.022		0.027		0.132*
		(0.029)		(0.029)		(0.066)
L5.(mean) wdi_trade		-0.000		-0.000		0.000*
		(0.000)		(0.000)		(0.000)
L5.(mean) natgini_		-0.161		-0.104		0.044
		(0.113)		(0.130)		(0.194)
L5.(mean) chprtpct		0.019		0.016		0.204*
		(0.078)		(0.082)		(0.113)
L5.(mean) bmr_dem		-0.001		0.001		0.000
		(0.011)		(0.011)		(0.027)
L5.(mean) press_freedom		0.001***		0.001***		0.001*
		(0.000)		(0.000)		(0.001)
Observations	2869	1094	2869	1094	539	208

Standard errors in parentheses

Thresholds: Strict90. Year FE included. L5. applied. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

No Mexico

As Mexico is an outlier in the ICEDI, these tables show the results of interactions with the ICEDI where Mexico is dropped from the data beforehand. The models are according to different time lags, as in previous regressions going from no time lag, to 1,3 and finally 5 years. The thresholds are according to the base model.

Table 38: Expenditure decentralisation, Mexico excluded

Robustness (No Mexico): expdec Across Lags

	(1)	(2)	(3)	(4)
	Current	Lag1	Lag3	Lag5
(mean) expdec	-0.625***			
	(0.147)			
(mean) expdec # (mean) ICEDI	0.061			
	(0.126)			
L.(mean) expdec		-0.279		
		(0.201)		
L3.(mean) expdec			-0.060	
			(0.104)	
L5.(mean) expdec				-0.271***
				(0.082)
L.(mean) expdec # L.(mean) ICEDI		0.074		
		(0.093)		
L3.(mean) expdec # L3.(mean) ICEDI			0.167**	
			(0.068)	
L5.(mean) expdec # L5.(mean) ICEDI				0.181*
				(0.096)
(mean) ICEDI	0.053			
	(0.055)			
L.(mean) ICEDI		-0.034		
		(0.058)		
L3.(mean) ICEDI			-0.113***	

			(0.039)	
L5.(mean) ICEDI				-0.068
				(0.044)
ln_gdp	-0.176**			
	(0.064)			
ln_pop	-0.165			
	(0.140)			
(mean) wdi_trade	-0.000			
	(0.001)			
(mean) natgini_	0.236			
	(0.387)			
(mean) chprtptct	0.578			
	(0.373)			
(mean) bmr_dem	-0.137**			
	(0.052)			
(mean) press_freedom	0.004*			
	(0.002)			
L.ln_gdp		-0.050		
		(0.094)		
L3.ln_gdp			0.039	
			(0.050)	
L5.ln_gdp				-0.043
				(0.049)
L.ln_pop		-0.138		
		(0.157)		
L3.ln_pop			-0.102	
			(0.110)	
L5.ln_pop				-0.084
				(0.078)
L.(mean) wdi_trade		-0.000		

		(0.001)		
L3.(mean) wdi_trade			0.000	
			(0.001)	
L5.(mean) wdi_trade				0.001***
				(0.000)
L.(mean) natgini_		-0.287		
		(0.305)		
L3.(mean) natgini_			0.127	
			(0.159)	
L5.(mean) natgini_				-0.212
				(0.277)
L.(mean) chprtpct		0.365		
		(0.372)		
L3.(mean) chprtpct			-0.364	
			(0.402)	
L5.(mean) chprtpct				-0.649*
				(0.351)
L.(mean) bmr_dem		-0.040		
		(0.068)		
L3.(mean) bmr_dem			-0.079*	
			(0.039)	
L5.(mean) bmr_dem				-0.104**
				(0.039)
L.(mean) press_freedom		0.001		
		(0.002)		
L3.(mean) press_freedom			0.002	
			(0.001)	
L5.(mean) press_freedom				0.002
				(0.001)
Observations	107	107	107	94

Standard errors in parentheses

Thresholds: Base (75/50). Mexico excluded. Year FE included. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 39: Political decentralisation, Mexico excluded

Robustness (No Mexico): h_f Across Lags

	(1)	(2)	(3)	(4)
	Current	Lag1	Lag3	Lag5
(mean) $h_f=0$	0.000			
	(.)			
(mean) $h_f=1$	0.047			
	(0.029)			
(mean) $h_f=0$ # (mean) ICEDI	0.000			
	(.)			
(mean) $h_f=1$ # (mean) ICEDI	-0.156***			
	(0.057)			
0bL.(mean) h_f		0.000		
		(.)		
1L.(mean) h_f		0.052*		
		(0.026)		
0bL.(mean) h_f # oL.(mean) ICEDI		0.000		
		(.)		
1L.(mean) h_f # L.(mean) ICEDI		-0.166***		
		(0.053)		
0bL3.(mean) h_f			0.000	
			(.)	
1L3.(mean) h_f			0.010	
			(0.022)	
0bL3.(mean) h_f # oL3.(mean) ICEDI			0.000	
			(.)	
1L3.(mean) h_f # L3.(mean) ICEDI			-0.276***	
			(0.047)	
0bL5.(mean) h_f				0.000

				(.)
1L5.(mean) h_f				0.016
				(0.018)
0bL5.(mean) h_f # oL5.(mean) ICEDI				0.000
				(.)
1L5.(mean) h_f # L5.(mean) ICEDI				-0.109**
				(0.045)
(mean) ICEDI	0.078**			
	(0.036)			
L.(mean) ICEDI		0.071*		
		(0.037)		
L3.(mean) ICEDI			0.083**	
			(0.031)	
L5.(mean) ICEDI				0.058
				(0.042)
ln_gdp	0.011			
	(0.051)			
ln_pop	0.495***			
	(0.151)			
(mean) wdi_trade	0.002**			
	(0.001)			
(mean) natgini_	-0.267			
	(0.247)			
(mean) chprtpct	-0.256			
	(0.435)			
(mean) bmr_dem	0.044			
	(0.063)			
(mean) press_freedom	0.002			
	(0.002)			
L.ln_gdp		0.027		

		(0.047)		
L3.ln_gdp			-0.031	
			(0.038)	
L5.ln_gdp				-0.041
				(0.029)
L.ln_pop		0.432***		
		(0.152)		
L3.ln_pop			0.218**	
			(0.092)	
L5.ln_pop				0.116*
				(0.064)
L.(mean) wdi_trade		0.001**		
		(0.001)		
L3.(mean) wdi_trade			0.001**	
			(0.000)	
L5.(mean) wdi_trade				0.000
				(0.000)
L.(mean) natgini_		-0.132		
		(0.241)		
L3.(mean) natgini_			-0.112	
			(0.183)	
L5.(mean) natgini_				0.033
				(0.176)
L.(mean) chprtpct		-0.096		
		(0.409)		
L3.(mean) chprtpct			0.093	
			(0.264)	
L5.(mean) chprtpct				0.222*
				(0.120)
L.(mean) bmr_dem		0.015		

		(0.045)		
L3.(mean) bmr_dem			0.034	
			(0.033)	
L5.(mean) bmr_dem				0.005
				(0.026)
L.(mean) press_freedom		0.002		
		(0.001)		
L3.(mean) press_freedom			0.001	
			(0.001)	
L5.(mean) press_freedom				0.001**
				(0.001)
Observations	205	205	205	206

Standard errors in parentheses

Thresholds: Base (75/50). Mexico excluded. Year FE included. Clustered SEs by Country.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

AI disclosure statement

Generative AI has been used to generate STATA code for the thesis and troubleshooting code. It has not been used to generate text nor theory for the thesis.

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